A Survey of On-Policy Distillation for Large Language Models

Mingyang Song & Mao Zheng
Large Language Model Department
Tencent, China
nickmysong@tencent.com

Abstract

Knowledge distillation has transitioned from a model compression technique into a general capability transfer engine for Large Language Models (LLMs). The dominant off-policy paradigm forces students to learn from static, teacher-generated trajectories, triggering exposure bias when students encounter their own distribution shifts during autoregressive generation. On-Policy Distillation (OPD) eliminates this mismatch by soliciting teacher feedback on the student's self-generated trajectories, transforming distillation into an interactive optimization loop. We present the first comprehensive survey dedicated to OPD for LLMs, covering over 90 methods published through April 2026 and abandoning descriptive enumeration in favor of a structured practitioner's decision chain. We formulate a unified mathematical framework for trajectory-based divergence minimization, connecting the f-divergence family (Forward KL, Reverse KL, JSD, α-divergence) to their information-geometric interpretations and practical implications. We then classify the rapidly growing literature not by heuristic boundaries, but by the core technical contribution of each method: objective function design, signal source architecture, and training dynamics optimization. By charting the transition from token-level logit matching to RL-augmented sequence objectives and self-play, we reveal the theoretical success conditions of OPD, catalog failure modes (exposure bias, self-play saturation, teacher unreliability), and chart future directions in scaling laws, uncertainty-aware feedback, and agentic distillation.

Awesome List & Paper Collection
https://github.com/nick7nlp/Awesome-LLM-On-Policy-Distillation

1 Introduction

Knowledge distillation, introduced by Hinton et al. (2015), compresses the knowledge of a large teacher network into a smaller student by training the student to match the teacher's softened output distribution. For autoregressive language models, this paradigm was extended to sequence-level objectives by Kim & Rush (2016b), who showed that matching the teacher's full sequence distribution outperforms word-level matching for machine translation. These foundational works established the template that dominated the field for nearly a decade: generate a dataset from the teacher, then train the student on this static corpus.

The emergence of Large Language Models (LLMs) has elevated distillation from a compression technique to the primary vector for capability transfer. DeepSeek-R1 (DeepSeek-AI et al., 2025) demonstrated that distilling chain-of-thought reasoning from a 671B MoE teacher can produce small models (1.5B–70B) that rival frontier systems on competition-level mathematics. However, this success masks a fundamental limitation: the student is trained on the teacher's trajectories, not its own.

Conventional distillation operates *off-policy*. The student model p_θ learns to replicate the teacher's next-token distribution over a static corpus p_{data}. During training, the student is teacher-forced, conditioning its predictions on flawless prefixes. During inference, it must generate autoregressively, conditioning on its own (often imperfect) outputs. This train-test discrepancy is a classic manifestation of *exposure bias* (Ross et al., 2011). When a student makes a slight error, the resulting prefix shifts away from the training manifold. Lacking supervisory signals for these self-induced out-of-distribution states, the student's errors compound, with the expected discrepancy scaling quadratically as $O(\epsilon T^2)$ over a trajectory of length T.

Gudibande et al. (2023) provided an influential early warning: imitating proprietary LLMs through off-policy distillation on API-generated data produces models that mimic surface style while failing to transfer genuine reasoning capability. This "false promise" of imitation learning motivated the search for training paradigms that force the student to confront its own weaknesses.

On-Policy Distillation (OPD) directly attacks exposure bias. It shifts the sampling distribution from the static dataset to the student's evolving policy. The student generates its own trajectories, explores sub-optimal states, and queries the teacher for corrective feedback on those specific states. This transitions distillation from supervised sequence modeling to an interactive reinforcement learning framework, recovering the theoretical $O(T)$ compounding error bounds of the DAgger algorithm (Ross et al., 2011; Agarwal et al., 2024).

The resulting literature is dense and highly fragmented. Methods drawing from imitation learning (GKD (Agarwal et al., 2024)), reinforcement learning (MiniLLM (Gu et al., 2024)), and self-play (SPIN (Chen et al., 2024)) often mask identical mathematical objectives behind disparate terminology. The field has grown explosively: while fewer than 10 papers addressed on-policy LLM distillation before 2024, over 90 methods have appeared in the subsequent 18 months, driven by both the practical demands of efficient model deployment and the theoretical realization that the boundary between distillation and reinforcement learning is dissolving. Industrial adoption has been equally rapid: Qwen3 (Yang et al., 2025a), DeepSeek-R1 (DeepSeek-AI et al., 2025), Gemma 2 (Gemma Team et al., 2024), and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) all incorporate OPD as a core training ingredient, transitioning the technique from research curiosity to production necessity.

Positioning relative to existing surveys. Previous surveys on knowledge distillation for LLMs (Xu et al., 2024) treat OPD as one subsection within a broader compression taxonomy, dedicating only a few pages to on-policy methods. General KD surveys cover teacher-student compression, feature matching, and data augmentation but do not provide the unified mathematical treatment or the practitioner-centric organization that this rapidly maturing subfield demands. Our survey differs in three ways: (1) we focus exclusively on on-policy methods, enabling a deeper analysis than any general survey can provide, (2) we organize methods by a practitioner's decision chain rather than by descriptive categories, directly addressing the "which method should I use?" question, and (3) we provide a unified f-divergence framework that reveals how seemingly disparate methods (GKD, MiniLLM, DistiLLM, G-OPD) instantiate the same underlying objective under different parameterizations.

We contribute the following:

1. **A Unified Formalism:** We construct a continuous mathematical framework mapping fixed divergences, adaptive divergences, and RL-augmented KD into a shared geometric space (Section 2).
2. **A Decision-Centric Taxonomy:** We replace fuzzy descriptive categories with a practitioner's decision tree (Section 3), organizing methods by their exact functional contribution: Objective Design (Section 4), Signal Source (Section 5), and Training Dynamics (Section 6).
3. **Theoretical Failure Analysis:** We synthesize empirical failures (flawed prefix traps, saturation) into a coherent understanding of when and why OPD breaks down (Section 7).

4. **Comprehensive Coverage:** We survey over 90 methods spanning white-box logit distillation, black-box API distillation, self-distillation, and RL-augmented objectives, with detailed comparison tables (Tables 1, 2, and 3).

5. **Actionable Guidance:** We provide a concrete practitioner decision tree (Section 3.3) that maps deployment constraints to recommended methods, compute budget guidelines, and a "When to Use On-Policy vs. Off-Policy" reference (Section 8.4).

Roadmap. Section 2 establishes mathematical preliminaries, formalizing autoregressive generation as a sequential decision problem and deriving the exposure bias that motivates on-policy approaches. Section 3 presents our practitioner-centric landscape and decision tree. Sections 4–6 provide an in-depth analysis of OPD methods organized by the three stages of the decision pipeline: objective function design, signal source architecture, and training dynamics optimization. The ordering of these sections is not arbitrary: each addresses a design decision that logically precedes the next, tracing the arc from "what to optimize" through "where the signal comes from" to "how to make training work in practice." Section 7 consolidates theoretical perspectives on success conditions, failure modes, and the on-policy vs. off-policy tradeoff. Section 8 discusses industrial deployments, system-level optimization, and emerging domains. Section 9 identifies open problems and future directions.

Paper statistics. This survey covers over 100 distinct methods from 2023–2026, drawn from top venues (ICLR, ICML, NeurIPS, EMNLP, ACL) and recent preprints. Approximately 60% of the surveyed papers appeared after January 2025, reflecting the field's rapid acceleration. We include 23 formal equations, 3 comprehensive comparison tables, a practitioner decision tree, and a complete experimental configuration reference spanning mathematical reasoning, code generation, instruction following, multimodal, and emerging domains.

Scope. We focus on methods where the student generates its own training data during the distillation process. We include white-box distillation (teacher logits accessible), black-box distillation (only teacher API or text outputs), self-distillation (no external teacher), and RL-augmented distillation (combining teacher supervision with reward signals). We exclude general knowledge distillation that uses only off-policy data, model compression techniques orthogonal to the training paradigm (pruning, quantization, low-rank factorization), and inference-time methods (self-consistency, majority voting, chain-of-thought prompting) that do not modify model weights. We also deliberately include methods from adjacent fields (imitation learning, online RL, preference optimization) when they operate on student-generated data with teacher or verifier supervision, as the boundary between these fields and OPD has effectively dissolved.

Notation convention. Throughout, p_T denotes the teacher distribution, p_θ the student parameterized by θ, π_{mix} the sampling policy for on-policy generation (which may differ from p_θ for mixed-policy methods), and $D_{\text{KL}}(P\|Q)$ the Kullback-Leibler divergence with the convention that P is the reference and Q is the approximation. When we write $y \sim p_\theta(\cdot|x)$, we mean the full autoregressive rollout generated token-by-token from the student's current policy.

2 Background and Unified Math

The necessity of OPD stems from the compounding geometry of autoregressive generation. This section formalizes the distribution mismatch, establishes the unified f-divergence framework that subsumes modern OPD objectives, and traces the historical arc from classical KD to modern on-policy methods.

Defining "on-policy." A distillation method is *on-policy* if the training data for the student is sampled from the student's own current policy p_θ at training time, rather than from a fixed external corpus $\mathcal{D}$ or from the teacher's generation distribution p_T. Formally, on-policy training optimizes:

$$\min_\theta \mathbb{E}_{x\sim\mathcal{D}}\, \mathbb{E}_{y\sim p_\theta(\cdot|x)}\left[\mathcal{L}(y, x; \theta, T)\right] \tag{1}$$

where $\mathcal{L}$ is the distillation loss (divergence, reward, or hybrid). The critical distinction is that the *outer* expectation is over the student's own generations, not a static dataset. This creates a non-stationary optimization landscape: as θ updates, the data distribution p_θ shifts, requiring fresh rollouts at each training step. The computational cost of this fresh generation is the fundamental systems-level challenge of OPD (Section 6.3).

Notation. Throughout this survey, we use lowercase p for token-level conditional distributions ($p_T(\cdot|x, y_{<t})$ for teacher, $p_\theta(\cdot|x, y_{<t})$ for student) and uppercase P for sequence-level distributions ($P_T(y|x)$, $P_\theta(y|x)$). $\mathcal{D}$ denotes the training dataset, p_{data} the data distribution, and $|V|$ the vocabulary size. D_{KL} and D_f denote KL divergence and general f-divergence respectively. In the RL formalization, π denotes policies and $R(x, y)$ the outcome reward. When discussing individual methods, we follow each paper's conventions where clarity demands.

2.1 Classical Knowledge Distillation

The original KD formulation (Hinton et al., 2015) trains the student to match the teacher's temperature-softened output distribution:

$$p_T^{(\tau)}(y|x) = \frac{\exp(z_y/\tau)}{\sum_j \exp(z_j/\tau)} \tag{2}$$

where z_y are the teacher's logits and $\tau > 1$ is a temperature parameter. When $\tau \to \infty$, the softened distribution approaches a uniform distribution over the vocabulary, revealing the teacher's "dark knowledge" about inter-class similarities that are hidden in the hard labels. The gradient of the distillation loss with respect to the student's logits z_i^S is proportional to $\frac{1}{\tau}(p_i^S - p_i^T)$, demonstrating that higher temperatures amplify the contribution of non-peak tokens, effectively transferring the teacher's relational structure between vocabulary items.

For autoregressive LLMs, Kim & Rush (2016b) extended this to Sequence-Level Knowledge Distillation (Seq-KD). Instead of factorizing the loss token-by-token over fixed dataset prefixes, Seq-KD minimizes the global KL divergence over the entire sequence space:

$$\mathcal{L}_{\text{Seq-KD}} = -\sum_{y \in \mathcal{Y}} P_T(y|x) \log P_\theta(y|x) \tag{3}$$

Since $|\mathcal{Y}| = |V|^T$ grows exponentially with length, this is intractable. Seq-KD approximates $P_T(y|x)$ with a Dirac delta at the beam-search output $\hat{y} = \arg\max_y P_T(y|x)$, reducing to standard negative log-likelihood on teacher-generated sequences. This approximation works well when the teacher is confident but discards all information about the teacher's uncertainty, a limitation that motivates the softer distributional methods below.

Subsequent work (Zhong et al., 2024) revisited these foundations for modern LLMs, demonstrating that classical assumptions (small teacher-student gap, shared vocabulary, similar architectural inductive biases) frequently fail in the era of heterogeneous model families. This motivates the more general f-divergence framework below.

From classical KD to on-policy. The historical progression from Hinton's classical KD to modern OPD can be understood as a series of relaxations. Classical KD assumes: (1) shared vocabulary, (2) i.i.d. data, (3) static teacher, (4) off-policy training data. Seq-KD (Kim & Rush, 2016a) relaxes assumption (2) by operating at the sequence level but retains the static data assumption. GKD (Agarwal et al., 2024) relaxes assumption (4) by training on student-generated sequences. DSKD (Zhang et al., 2025b) relaxes assumption (1) through dual-space alignment. G-OPD (Yang et al., 2026c) relaxes all four simultaneously while adding RL-augmented objectives. Each relaxation addresses a specific failure mode that becomes more severe as model scale increases: vocabulary mismatch is irrelevant for same-family distillation but catastrophic for cross-family (relaxation 1), while exposure bias is negligible for short sequences but devastating for multi-step reasoning (relaxation 4). This progression is not merely historical but prescriptive: practitioners should identify which assumptions hold in their specific setting and select the minimally relaxed method, avoiding unnecessary complexity.

2.2 Off-Policy Exposure Bias

Standard Knowledge Distillation minimizes the KL divergence over states drawn from the dataset distribution $d_{\mathcal{D}}(s)$:

$$\mathcal{L}_{\text{Off-Policy}} = \mathbb{E}_{x,y \sim p_{\text{data}}} \left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_T(\cdot|x, y_{<t}) \parallel p_\theta(\cdot|x, y_{<t})) \right] \tag{4}$$

At inference, the student acts according to its own policy, inducing a different state visitation distribution $d_{\pi_\theta}(s)$. By the DAgger theorem (Ross et al., 2011), if a policy mimics an expert with per-step error ϵ under the training distribution, the expected total discrepancy over a trajectory of length T under the student's own distribution scales quadratically, bounded by $O(\epsilon T^2)$.

This quadratic compounding has severe practical consequences for reasoning tasks. A mathematical proof requiring 10 reasoning steps with per-step accuracy 95% yields only $(0.95)^{10} \approx 60\%$ trajectory-level correctness under off-policy training, because the student never learns to recover from its own intermediate errors. On-policy training exposes the student to its own error states during training, enabling it to learn correction strategies that reduce the effective per-step error rate when errors have already occurred.

Remark: The DAgger bound in LLMs. While the theoretical transition from $O(\epsilon T^2)$ to $O(\epsilon T)$ is appealing, applying the DAgger bound to LLMs requires nuance. The original theorem assumes an *interactive expert* providing optimal actions in any state. In white-box OPD, the "expert" provides a next-token distribution $p_T(y_t|\hat{y}_{<t})$ conditioned on the student's prefix $\hat{y}_{<t}$. If the student hallucinates a severely out-of-distribution prefix, the teacher's conditional distribution may itself become poorly calibrated, because the teacher was never trained on such inputs. Forcing the student to match this noisy distribution violates the core assumption of interactive imitation learning, potentially destabilizing training rather than recovering the $O(\epsilon T)$ bound. This observation provides the theoretical motivation for the adaptive trust mechanisms surveyed in Section 6.1 and explains why unconditional token-level matching is insufficient for robust OPD.

2.3 The Unified f-Divergence Framework

OPD resolves exposure bias by altering the training expectation to sample from the student's own rollouts, or a mixture policy π_{mix}. The generalized on-policy objective decouples the sampling trajectory from the local matching metric:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{y \sim \pi_{\text{mix}}} \left[\sum_{t=1}^{|y|} \mathcal{D}_f \left(p_T(\cdot|x, y_{<t}),\ p_\theta(\cdot|x, y_{<t}) \right) \right] \tag{5}$$

Here, $\mathcal{D}_f$ represents a divergence from the f-divergence family (Wen et al., 2023). Formally, given two distributions P and Q, an f-divergence is defined as:

$$D_f(P \parallel Q) = \mathbb{E}_{y \sim Q} \left[f\left(\frac{P(y)}{Q(y)} \right) \right] \tag{6}$$

where $f : (0, \infty) \to \mathbb{R}$ is a convex generator with $f(1) = 0$. The choice of f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$:

- **Forward KL** ($f(u) = u \log u$): Mode-covering (zero-avoiding). The student must allocate probability mass wherever the teacher does, often bridging distinct teacher modes and hallucinating in the inter-mode space.
- **Reverse KL** ($f(u) = -\log u$): Mode-seeking (zero-forcing). The student collapses its mass onto the teacher's single highest peak, ignoring secondary acceptable outputs but guaranteeing high precision.
- **JSD** ($f(u) = u \log u - (u+1) \log \frac{u+1}{2}$): Symmetric, bounded, and smoothly interpolates between mode-covering and mode-seeking behavior.

- α**-divergence**: Parameterized family that continuously interpolates between Forward KL ($\alpha \to 1$) and Reverse KL ($\alpha \to 0$), enabling fine-grained control over the mode-seeking/covering tradeoff.

The practical implications of this taxonomy are profound. For tasks with a *unique* correct answer (mathematical proofs, code correctness), Reverse KL's mode-seeking behavior concentrates the student on the single optimal solution path. For tasks with *many* acceptable outputs (creative writing, open-ended QA), Forward KL's mode-covering behavior preserves output diversity. JSD offers a compromise but lacks theoretical optimality guarantees for either extreme. The key theoretical insight is that all these divergences are computable as expectations under the student's own policy (since $D_f(P_T \parallel P_\theta) = \mathbb{E}_{y\sim p_\theta}[f(p_T(y)/p_\theta(y))]$), which is precisely why they are amenable to on-policy optimization via policy gradient methods.

This framework subsumes all modern OPD objectives: the choice of f determines the geometric behavior, while the choice of π_{mix} controls the degree of on-policy exploration. We illustrate by mapping the three foundational methods into this space:

GKD (Agarwal et al., 2024) defines $\pi_{\text{mix}} = \lambda p_\theta + (1-\lambda)p_{\text{data}}$ as an explicit interpolation, and is agnostic to $\mathcal{D}_f$, empirically testing Forward KL, Reverse KL, and JSD. Setting $\lambda \to 1$ makes GKD purely on-policy, while $\lambda = 0$ reduces to standard off-policy KD. The λ parameter provides a smooth dial between the extremes: moderate values ($\lambda \approx 0.5$) provide the benefits of on-policy exposure while retaining the stability of grounded, dataset-derived prefixes. GKD's experiments establish that on-policy sampling ($\lambda = 1$) with JSD divergence yields the best downstream performance on summarization and translation, empirically validating the exposure bias hypothesis.

MiniLLM (Gu et al., 2024) selects Reverse KL as $\mathcal{D}_f = D_{\text{KL}}(p_\theta \parallel p_T)$ and employs $\pi_{\text{mix}} = (1-\alpha)p_\theta + \alpha p_T$ with $\alpha = 0.2$ for stability. Because Reverse KL places the student in both the expectation and the log-ratio, MiniLLM reformulates optimization via REINFORCE, treating $\log(p_T/p_\theta)$ as a per-step reward.

DistiLLM (Ko et al., 2024) maintains $\pi_{\text{mix}} = p_\theta$ (fully on-policy) but replaces the KL target from p_T to a skewed mixture $\tilde{p} = \alpha p_T + (1-\alpha)p_\theta$, computing $D_{\text{KL}}(p_T \parallel \tilde{p})$. This avoids zero-division instability when $p_\theta(y) \approx 0$, because $\tilde{p}(y) \geq \alpha p_T(y) > 0$ always, achieving tractability and stability without policy gradients.

This decomposition reveals that modern OPD is not a series of disparate heuristics but a systematic exploration of a design space governed by three interacting choices: trajectory sampling (π_{mix}), divergence generator (f), and argument ordering within the chosen divergence.

2.4 Distillation Scaling Laws

A critical gap in the field is the absence of scaling laws specific to distillation. While Hoffmann et al. (2022) established compute-optimal scaling for pre-training (the Chinchilla laws), and Busbridge et al. (2025) initiated the study of distillation-specific scaling, no comprehensive framework yet predicts how distillation quality scales with teacher size, student size, data volume, and on-policy rollout budget. This absence forces practitioners to rely on expensive trial-and-error. Preliminary evidence suggests that distillation scaling obeys qualitatively different laws from pre-training: the student benefits from teacher scale even after the teacher's marginal pre-training gains plateau, because the teacher's improved calibration (better next-token probability estimates) provides denser gradient signal per training step. Furthermore, on-policy rollout budget appears to exhibit a log-linear relationship with downstream quality once sufficient baseline capability is established, with roughly 2× rollout budget yielding consistent 2–3% absolute improvements on reasoning benchmarks regardless of student scale. However, these observations remain empirical rather than predictive, and the theoretical frameworks developed in Section 7 provide only partial answers. A complete distillation scaling law, expressed as Quality $\propto N_T^\alpha \cdot N_S^\beta \cdot D^\gamma \cdot R^\delta$ where R is the rollout budget, remains an open problem (Section 9).

3 Landscape and Practitioner Guide

To navigate the fragmented OPD literature, we construct a functional landscape. Methods are classified not by arbitrary labels, but by the specific step in the engineering pipeline they optimize.

3.1 Method Landscape

The practitioner faces three sequential decisions: (1) what objective to optimize, (2) where the supervisory signal originates, and (3) how to stabilize the resulting training dynamics. Figure 1 illustrates this pipeline.

Objective Design (§4)	$\longrightarrow$	**Signal Source (§5)**	$\longrightarrow$	**Training Dynamics (§6)**
Fixed/Adaptive/RL		White-Box/Black-Box/Self		Weighting/Curriculum/Efficiency

Figure 1: The functional landscape of On-Policy Distillation. The taxonomy reflects the practitioner's decision chain: first choose the objective, then identify the signal source, then stabilize the training loop.

The three-stage pipeline is not merely organizational convenience: it reflects the actual engineering workflow. A practitioner first chooses an objective function (Forward KL? Reverse KL? Adaptive? RL-augmented?), then selects a signal source (white-box logits, black-box API, or self-generated), and finally addresses the training dynamics challenges that arise from the chosen combination. Some combinations are incompatible (e.g., Forward KL requires white-box logits, eliminating all black-box signal sources), which our decision tree (Section 3.3) makes explicit. Other combinations are synergistic: RL-augmented objectives (G-OPD) naturally pair with external feedback signal sources (verifiers, reward models), while fixed divergences pair most naturally with white-box sources where the full distribution is available for exact computation.

3.2 Method Comparison Table

3.3 Practitioner Decision Tree

The following decision tree provides actionable guidance for selecting OPD methods based on access constraints, task requirements, and compute budgets. These recommendations are derived from the empirical patterns observed across the methods surveyed in Sections 4–6.

1. **Do you have full logit access to the teacher?**
   - *Yes (White-Box):* Use DistiLLM (Ko et al., 2024) for stable token-level matching on instruction following, or G-OPD (Yang et al., 2026c) if strong reasoning extrapolation is required. For adaptive divergences, ToDi (Jung et al., 2025) or Entropy-Aware (Jin et al., 2026) provide automatic switching. If teacher and student use different tokenizers, deploy DSKD (Zhang et al., 2025b) or Cross-Tokenizer KD (Boizard et al., 2025).
   - *No, API only (Black-Box):* Use OVD (Xiong et al., 2026) for verbal score distillation (simplest), GAD (Ye et al., 2025) for adversarial training (strongest signal), or DAIL (Mendes et al., 2026) for inverse-RL framing. If only text outputs are available and the student is small, LUFFY (Yan et al., 2025) with mixed-policy GRPO provides a strong baseline.
   - *No teacher available (Self-Distillation):* If oracle answers are available, use Privileged Information methods: OPSD (Zhao et al., 2026b) for math reasoning, GATES (Stein et al., 2026) for open-domain QA. If no oracle is available, start with SD-ZERO (He et al., 2026) (requires only a binary verifier) or SSD (Zhang et al., 2026d) (requires nothing at all, but saturates quickly).
2. **What is the task profile?**

Classification methodology. Each method is assigned to exactly one primary category based on its *core contribution dimension*: if a paper's novelty lies in the loss function (e.g., G-OPD's KL-constrained RL reformulation), it belongs to "Objective." If it introduces a novel signal architecture (e.g., DSKD's dual-space projection), it belongs to "Signal." If it solves a training stability or efficiency problem (e.g., TIP's token weighting), it belongs to "Dynamics." Methods that contribute to multiple dimensions are classified by their *most distinctive* contribution. This strict one-method-one-category rule prevents the overlap and confusion that plagued previous multi-dimensional taxonomies.

Table 1: Comparison of key OPD methods categorized by their primary contribution dimension.

Method	Category	Core Contribution	Signal Source	Loss Granularity
GKD (Agarwal et al., 2024)	Objective (Fixed)	DAgger for LLMs; interpolation	White-Box Logits	Token
DistiLLM (Ko et al., 2024)		Skewed KL mixtures	White-Box Logits	Token
MiniLLM (Gu et al., 2024)		Sequence-level Reverse KL via RL	White-Box Logits	Sequence
ToDi (Jung et al., 2025)	Objective (Adaptive)	Entropy-routed token divergence	White-Box Logits	Token
Entropy-Aware (Jin et al., 2026)		Entropy-based KL interpolation	White-Box Logits	Token
G-OPD (Yang et al., 2026c)	Objective (RL)	OPD equivalence to KL-RL	White-Box Logits	Sequence/Token
RLKD (Xu et al., 2025b)		KD augmented with reward model	Reward Model	Sequence
DSKD (Zhang et al., 2025b)	Signal (White-Box)	Cross-architecture alignment	White-Box Logits	Token
Delta-KD (Ko et al., 2026)		Base-to-Instruct delta isolation	White-Box Logits	Token
Lion (Jiang et al., 2023)	Signal (Black-Box)	Verbal feedback curriculum	Black-Box API	Sequence
GAD (Ye et al., 2025)		Adversarial distribution matching	Black-Box API	Sequence
OPSD (Zhao et al., 2026b)	Signal (Self)	Privileged Info (PI) conditioning	Self-Generated	Token
SPIN (Chen et al., 2024)		Self-play vs reference responses	Self-Generated	Sequence
TIP (Xu et al., 2026b)	Dynamics (Weight)	Token importance profiling via diff	White-Box Logits	Token
PACED (Xu et al., 2026a)		Beta-kernel curriculum pacing	White-Box Logits	Token

- *Reasoning / Math:* Mode-seeking objectives are mandatory to prevent hallucination. Use Reverse KL (MiniLLM (Gu et al., 2024)) or RL-augmented KD (G-OPD, KDRL (Xu et al., 2025a)). Combine with curriculum pacing (PACED (Xu et al., 2026a)) for compute efficiency.
- *Open-ended Generation:* Mode-covering objectives preserve diversity. Use Forward KL or adaptive entropy-aware divergences. Avoid aggressive Reverse KL, which collapses creative diversity.
- *Instruction Following:* GKD (Agarwal et al., 2024) with $\lambda \geq 0.5$ provides the best quality-compute tradeoff. AlignDistil (Zhang et al., 2025a) adds preference alignment.
- *Multi-step Agentic Tasks:* Requires sequence-level objectives with trajectory-aware credit assignment. SCoRe (Lyu et al., 2025) for error correction, RLAD (Zhang et al., 2026g) for trust-region stability.

3. **What are the compute limits?**
   - *Low compute (<500 GPU-hours):* Standard GKD, Fast OPD (Zhang et al., 2026a) (prefix truncation), or Lightning OPD (Wu et al., 2026) (4× cost reduction via offline caching).
   - *Medium compute:* Full on-policy training with adaptive divergences (ToDi, AKL (Wu et al., 2025)). Add PACED for efficient sample selection.
   - *High compute (>5000 GPU-hours):* Multi-stage pipelines: off-policy warmup → on-policy refinement → RL exploration. This is the recipe used by Qwen3 (Yang et al., 2025a) and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026).
4. **Stability concerns?**
   - *Length inflation:* Apply Stable-OPD (Luo et al., 2026) stabilization (reference divergence + rollout mixing).
   - *Flawed prefix trap:* Deploy TIP (Xu et al., 2026b) or SCOPE (Zheng et al., 2026) for teacher reliability filtering.

- *Thinking-pattern mismatch:* Start with off-policy cold start (Li et al., 2026e), then transition to on-policy after initial distribution alignment.

4 Objective Functions and Optimization

The foundational algorithmic decision in OPD is the choice of objective function: what mathematical quantity does the student optimize at each training step? This choice governs gradient geometry, mode coverage, and ultimately the ceiling of achievable performance. The field has evolved through three distinct phases: fixed divergences that apply a single metric uniformly, adaptive divergences that modulate the metric per-token based on local distributional geometry, and RL-augmented objectives that break through the teacher's capability ceiling entirely. We trace this progression below, emphasizing the mathematical relationships that connect seemingly disparate methods.

4.1 Fixed Divergence Objectives

GKD (Agarwal et al., 2024) established the canonical OPD framework by adapting the DAgger algorithm (Ross et al., 2011) for autoregressive language models. It samples trajectories from a mixture policy $\pi_{\text{mix}} = \lambda p_\theta + (1-\lambda)p_{\text{data}}$ and applies standard f-divergences (Forward KL, Reverse KL, or JSD) at each token position. The mixture coefficient λ interpolates between fully off-policy ($\lambda = 0$) and fully on-policy ($\lambda = 1$), providing a single knob for exposure bias mitigation. On instruction-following and summarization benchmarks, even moderate on-policy mixing ($\lambda \geq 0.5$) consistently outperforms pure off-policy training, validating the distributional alignment hypothesis.

However, pure KL divergences exhibit numerical instability during on-policy exploration. When the student generates tokens to which the teacher assigns near-zero probability, Forward KL produces unbounded gradients. Conversely, Reverse KL ignores these regions entirely, creating blind spots in the loss landscape. DistiLLM (Ko et al., 2024) addresses this through *Skewed KL* (SKL), constructing a smoothed target $\tilde{p} = \alpha p_T + (1-\alpha)p_\theta$:

$$\mathcal{L}_{\text{SKL}} = \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} D_{\text{KL}}\big(p_T(\cdot|y_{<t}) \parallel \alpha p_T(\cdot|y_{<t}) + (1-\alpha)p_\theta(\cdot|y_{<t})\big)\right] \tag{7}$$

Skewing guarantees the target density is bounded below by $(1-\alpha)p_\theta$, eliminating gradient explosion without requiring RL policy gradients. DistiLLM further introduces Skewed Reverse KL (SRKL) for mode-seeking applications, creating a symmetric pair of stabilized divergences. Its successor, DistiLLM-2 (Ko et al., 2025), recognizes that teacher-generated and student-generated data carry fundamentally different learning signals and applies *asymmetric* objectives: Forward SKL on teacher-generated data (where mode-covering teaches the student to explore the teacher's full distribution) and Reverse SRKL on student-generated data (where mode-seeking teaches the student to sharpen its own best responses). This source-aware asymmetry yields consistent improvements over symmetric single-divergence baselines across instruction following and summarization tasks, empirically validating the theoretical insight that the optimal divergence is data-source-dependent.

The fixed-divergence family faces a fundamental limitation: the choice between Forward KL (mode-covering) and Reverse KL (mode-seeking) is a global decision applied uniformly across all token positions. Yet the teacher's distributional structure varies dramatically within a single sequence. At a mathematical operator token, the teacher concentrates probability mass on one or two correct symbols. At a conjunction or filler word, dozens of tokens share comparable probability. Applying a single divergence to both positions necessarily compromises: Forward KL wastes gradient signal on high-entropy positions where any token is acceptable, while Reverse KL discards useful coverage information at low-entropy positions where multiple valid continuations exist.

Sequence-level objectives offer a complementary perspective. MiniLLM (Gu et al., 2024) elevates Reverse KL to the full sequence level, where the student p_θ appears inside the sampling expectation. The central challenge is computing the gradient of an expectation over

p_θ when p_θ itself is parameterized by θ. Starting from $D_{\mathrm{KL}}(p_\theta \parallel p_T) = \mathbb{E}_{y\sim p_\theta}[\log p_\theta(y|x) - \log p_T(y|x)]$ and applying the log-derivative trick:

$$\nabla_\theta \mathcal{L}_{\mathrm{MiniLLM}} = \mathbb{E}_{y\sim p_\theta}\left[\Big(\log p_\theta(y|x) - \log p_T(y|x)\Big)\nabla_\theta \log p_\theta(y|x)\right] \tag{8}$$

This reveals that minimizing sequence-level Reverse KL is mathematically equivalent to policy gradient RL with reward $r(x, y) = \log p_T(y|x) - \log p_\theta(y|x)$. MiniLLM decomposes this into per-step rewards r_t and future returns $R_t = \sum_{t'=t}^{|y|} r_{t'}$, enabling variance-reducing baselines:

$$\nabla_\theta \mathcal{L}_{\mathrm{MiniLLM}} = -\mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|}(R_t - b)\,\nabla_\theta \log p_\theta(y_t|y_{<t})\right] \tag{9}$$

where b is a learned baseline. This places optimization squarely in the RL framework, with the teacher's log-probability serving as a dense reward signal. The connection to policy optimization explains MiniLLM's strong performance on reasoning tasks where sequence-level coherence matters more than token-level accuracy. However, REINFORCE estimation in massive combinatorial output spaces demands substantially more training iterations and sophisticated baseline subtraction to converge.

KETCHUP (Fan et al., 2025) refines this approach by replacing the single-sample REINFORCE estimator with a K-step truncated return, reducing gradient variance while preserving the sequence-level objective. The constrained KD framework of Zimmer et al. (2025) augments the sequence-level Reverse KL with a hard KL constraint implemented via state-augmented CMDP, ensuring that the student's divergence from the teacher remains bounded even during aggressive exploration.

Constrained optimization for stability. To combat the high variance of REINFORCE, Zimmer et al. (2025) recast distillation as a Constrained Markov Decision Process (CMDP). Instead of treating the KL penalty as a soft Lagrangian term with manually tuned λ, they maximize task reward subject to a hard constraint $D_{\mathrm{KL}}(p_\theta \parallel p_T) \leq \epsilon$. KETCHUP (Fan et al., 2025) addresses the variance problem from a different angle by inducing a K-step return via the Bellman Optimality Equation, replacing the high-variance single-step REINFORCE estimator with a multi-step formulation that provably reduces gradient variance, yielding more stable RL-based distillation especially for larger student models.

The token-level vs. sequence-level tradeoff. The choice between token-level and sequence-level objectives represents a fundamental bias-variance tradeoff. Token-level methods (GKD, DistiLLM) provide $|V|$-dimensional gradient signals at every position, yielding low variance but potentially biased gradients when the teacher's token-level distribution is poorly calibrated (the flawed prefix trap of Section 6.1). Sequence-level methods (MiniLLM, KETCHUP) provide unbiased gradients through the policy gradient theorem but suffer from high variance due to single-sample REINFORCE estimation over exponentially large sequence spaces. The practical resolution is task-dependent: for instruction following where token-level teacher quality is high, GKD-style methods dominate. For reasoning tasks where sequence-level coherence is paramount, MiniLLM-style methods are preferred despite their variance, and KETCHUP's truncated returns offer an intermediate point on this spectrum.

4.2 Adaptive Divergence Objectives

The limitations of fixed divergences motivate *adaptive* methods that select or interpolate divergences based on local distributional geometry. The core insight is that the optimal divergence at each token position depends on the teacher's local confidence: when the teacher is certain (low entropy), mode-seeking Reverse KL preserves precision, and when the teacher is uncertain (high entropy), mode-covering Forward KL preserves diversity.

ToDi (Jung et al., 2025) operationalizes this by routing each token through either Forward KL or Reverse KL based on the teacher's entropy $H(p_T(\cdot|y_{<t}))$:

$$\mathcal{L}_{\mathrm{ToDi}} = \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} 1[H_t < \tau]\cdot D_{\mathrm{KL}}(p_\theta \parallel p_T) + 1[H_t \geq \tau]\cdot D_{\mathrm{KL}}(p_T \parallel p_\theta)\right] \tag{10}$$

where τ is a learned or fixed entropy threshold. Entropy-Aware OPD (Jin et al., 2026) replaces the hard routing with a continuous interpolation coefficient $\alpha_t \propto H_t$, yielding a smooth blend:

$$\mathcal{L}_{\text{EA}} = \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|}(1-\alpha_t)D_{\text{KL}}(p_\theta \parallel p_T) + \alpha_t D_{\text{KL}}(p_T \parallel p_\theta)\right] \tag{11}$$

Both methods consistently outperform any fixed divergence on math reasoning (MATH, GSM8K) and open-ended generation (AlpacaEval), confirming that the optimal divergence is indeed position-dependent.

AKL (Wu et al., 2025) takes a more aggressive approach, dynamically interpolating between Forward KL and Reverse KL based on a per-token confidence metric derived from the teacher-student probability ratio. Unlike entropy-based routing, which depends only on the teacher's distributional shape, AKL considers the *relative* geometry of teacher and student, adapting more aggressively at positions where the gap is largest.

These adaptive methods admit a natural interpretation through information geometry. The Fisher information metric on the statistical manifold of autoregressive distributions varies across token positions. At reasoning-critical positions, the manifold is highly curved (the teacher concentrates probability on few tokens), making Reverse KL's mode-seeking behavior locally appropriate because it follows the manifold's natural gradient. At generation-flexible positions, the manifold is nearly flat (many synonyms are equally valid), and Forward KL's mode-covering behavior preserves useful entropy. Adaptive divergences implicitly navigate this varying curvature by selecting the locally optimal metric, which explains their consistent empirical superiority over any fixed choice.

The practical implication is clear: the field has moved decisively from "which divergence?" to "which divergence *where*?" The evidence from ToDi, AKL, and Entropy-Aware OPD converges on a single recommendation: no fixed divergence is ever optimal, and the simplest adaptive routing (even a hard entropy threshold) consistently outperforms the best-tuned fixed alternative.

4.3 RL-Augmented Objectives

Divergence matching, whether fixed or adaptive, intrinsically limits the student to the teacher's capability ceiling: perfect optimization yields a student that matches (but never exceeds) the teacher's distribution. RL-augmented objectives break this ceiling by introducing an external performance signal that can steer the student toward solutions the teacher would never produce.

OPD as KL-constrained RL. G-OPD (Yang et al., 2026c) establishes a formal equivalence between standard OPD and dense KL-constrained reinforcement learning. It formulates distillation as maximizing a token-level reward derived from teacher logits, penalized by divergence from a reference policy:

$$\max_\theta \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|}\alpha \log p_T(y_t|y_{<t}) - D_{\text{KL}}\big(p_\theta(\cdot|y_{<t}) \parallel p_{\text{ref}}(\cdot|y_{<t})\big)\right] \tag{12}$$

When $\alpha = 1$, this reduces to standard Reverse KL distillation. When $\alpha > 1$, the objective forces the student to *extrapolate* beyond the teacher's probability mass, enabling a phenomenon documented as *Reward Extrapolation*: the student discovers novel, structurally sound reasoning paths in regions where the teacher's generation probability $p_T(y|x)$ is low but the outcome reward remains high. On competition-level math, a 4B student trained with G-OPD surpasses a 30B teacher, demonstrating that OPD need not be a lossy compression step.

This equivalence has a deeper theoretical consequence: it bridges the distillation and RL communities, providing a shared analytical language. Any advance in KL-constrained RL (better trust regions, adaptive penalty coefficients, variance reduction) immediately transfers to OPD, and vice versa.

Gradient decomposition and the KD+RL complementarity. The unified KD+RL framework of Li et al. (2025) decomposes the hybrid gradient into two orthogonal components: a dense KD gradient for token-level imitation and a Monte Carlo RL gradient for reward maximization. The KD component dampens the high variance inherent in policy gradient estimation, while the RL component prevents the student from collapsing onto teacher modes that are suboptimal for the downstream task. REOPOLD (Li et al., 2025) further formalizes this decomposition, splitting each parameter update into an imitation vector (pulling toward the teacher) and an exploration vector (pushing toward higher trajectory returns), making the complementarity explicit and tunable.

Dense KD + sparse reward. RLKD (Xu et al., 2025b) exemplifies the integration of dense teacher supervision with sparse outcome feedback. It introduces a Generative Structure Reward Model (GSRM) that scores reasoning paths for structural coherence, combining token-level KL regularization with trajectory-level reward:

$$\max_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{x\sim\mathcal{D}, y\sim p_\theta(\cdot|x)} \left[R(x,y) - \beta D_{\mathrm{KL}}\big(p_\theta(\cdot|x) \parallel p_T(\cdot|x)\big)\right] \tag{13}$$

The mathematical structure explains why the combination works: the KL term provides analytically computable dense gradients, while the reward term contributes Monte Carlo policy gradients for outcome-driven exploration.

Joint vs. sequential optimization. A persistent engineering challenge is *how* to combine KD and RL within a training pipeline. Sequential approaches (distill-then-RL or RL-then-distill) suffer from catastrophic interference: the RL stage erases distilled knowledge through policy drift, while the KD stage suppresses novel solutions discovered through exploration. KDRL (Xu et al., 2025a) operationalizes joint optimization by adding an on-policy KL regularizer between student and teacher *during* RL training. RLAD (Zhang et al., 2026g) refines this with Trust Region Ratio Distillation (TRRD), replacing the standard KL regularizer with a PPO-style likelihood-ratio objective anchored to a teacher-old-policy mixture. This selective strategy follows the teacher only when its signal improves the policy update and ignores it otherwise, a principled solution to the problem that blind KL minimization can drag the student toward suboptimal trajectories on problems the teacher itself struggles with.

Preference-based objectives. When logit access is insufficient to capture global intent, several methods map the problem onto Direct Preference Optimization (DPO). AlignDistil (Zhang et al., 2025a) constructs synthetic preference pairs from student rollouts scored by the teacher, applying both standard DPO and reverse DPO objectives for alignment. OVD (Xiong et al., 2026) queries the teacher for pairwise preferences over student-generated responses, avoiding the need for continuous probability extraction entirely.

Targeted correction vs. wholesale imitation. The RL-augmented methods above still provide uniform supervision across the entire trajectory. A distinct failure mode emerges from this uniformity: when the student makes a single error in step 3 of a 10-step proof, wholesale imitation retrains all 10 steps, diluting the corrective gradient. SuperCorrect (Yang et al., 2025b) introduces hierarchical thought templates that scaffold the student's reasoning at both strategic and step-level granularity, then applies cross-model DPO to teach error localization. SCoRe (Lyu et al., 2025) pushes targeting to its logical extreme: rather than providing complete correction traces, the teacher identifies and corrects *only the earliest error* in the student's trajectory, concentrating all supervision at the single point of maximum leverage. This approach exploits a structural property of reasoning chains: errors cascade, so correcting the root cause eliminates all downstream consequences simultaneously.

From signal matching to environment reconstruction. The most radical extension asks whether the student can recover the teacher's *underlying optimization landscape*. $\mathcal{X}$-KD (Cai & Yuan, 2026) adopts the AVRIL framework to jointly model the teacher's latent reward function and perform policy distillation, encouraging the student to optimize in a reconstructed version of the teacher's training environment rather than simply mimicking its outputs. TSD-KD (Kim & Baek, 2026) combines indirect distillation (student generates candidates that the teacher re-ranks via preference feedback) with direct distillation (KL matching applied selectively to high-entropy tokens where teacher-student gaps are largest). These methods represent the current frontier: moving from matching the teacher's *behavior* to understanding its *incentives*.

The progression across objective types reveals a fundamental insight. Fixed divergences provide stable, well-understood optimization but are limited by the teacher's capability ceiling. Adaptive divergences improve efficiency by matching the local geometry of the loss landscape. RL-augmented objectives break through the ceiling entirely but introduce variance and instability. The optimal choice depends on the specific application: for instruction following where teacher quality is high, fixed or adaptive divergences suffice. For reasoning tasks where the student must exceed the teacher, RL augmentation is essential. For tasks where the teacher's training process is accessible, environment reconstruction offers the most complete knowledge transfer.

5 Signal Source and Teacher Architecture

The second major decision in the OPD pipeline concerns the origin and nature of the supervisory signal. Given a fixed objective function (Section 4), the practitioner must determine *where the teacher signal comes from*: full logit access (white-box), API-only text outputs (black-box), or the model's own internal asymmetries (self-distillation). Each signal source imposes distinct constraints on the applicable objectives and achievable performance, creating a structured design space that this section maps comprehensively.

The evolution of signal sources mirrors the broader trend toward accessibility and self-sufficiency. Early OPD methods (GKD, DistiLLM) assumed full white-box access to model internals, a requirement that limits distillation to same-organization deployments. The rise of proprietary API-only models (GPT-4, Claude) necessitated black-box methods that can extract knowledge from text outputs alone. Most recently, self-distillation methods eliminate the external teacher entirely, enabling continuous improvement without access to any stronger model. This progression, from full access through limited access to no external access, represents increasing autonomy at the cost of signal density, creating a fundamental trade-off that practitioners must navigate.

Complex reasoning is where OPD proves most indispensable and where off-policy methods encounter their hardest limits. The reason is structural: reasoning is *path-dependent*. A proof that begins "assume for contradiction..." requires entirely different subsequent tokens than one beginning "we proceed by induction..." If the student has never generated the inductive approach during training, it cannot recover at inference time when that path would be optimal. Off-policy training over a fixed dataset of teacher reasoning traces can only cover a finite set of solution paths, while the student needs to navigate its own unique landscape of partial solutions, dead ends, and recovery strategies. This insight motivates the on-policy formulation for reasoning transfer: the student generates its own intermediate reasoning steps $r^S \sim P_\theta(\cdot|x)$, and the teacher evaluates these student-generated chains rather than dictating its own:

$$\mathcal{L}_{\text{CoT-OPD}}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\, r^S\sim P_\theta(\cdot|x)}\left[\sum_{t=1}^{|r^S|} D_{\text{KL}}\Big(P_T(\cdot|x, r^S_{<t}) \parallel P_\theta(\cdot|x, r^S_{<t})\Big)\right] \tag{14}$$

The Reverse KL direction is mode-seeking by design: rather than forcing the student to spread probability across all teacher-approved continuations (which would blur distinct reasoning strategies), it reinforces the student's own high-probability paths that the teacher validates. This self-consistent learning dynamic, where the student practices its own reasoning style under expert supervision, is why on-policy CoT distillation has become the de facto standard for instilling multi-step reasoning into smaller models (Hsieh et al., 2023).

5.1 White-Box Logit Supervision

White-box access provides the densest possible signal: the full probability distribution across $|V|$ tokens at every decoding step. This exposes the teacher's implicit decision boundaries, including the relative probabilities among incorrect tokens (Hinton's "dark knowledge"), which carry information about similarity structure that binary labels cannot capture.

Standard logit distillation. The majority of methods surveyed in Section 4 assume white-box access. GKD, DistiLLM, ToDi, G-OPD, and their variants all require computing $p_T(\cdot|x, y_{<t})$ at each student-generated token position. The computational cost is dominated by the teacher forward pass, which must be executed once per student-generated token (or once per sequence for cached implementations).

Token-level adaptive supervision. Within the white-box setting, several methods refine *which* tokens receive supervision. SelecTKD (Huang et al., 2025) applies confidence-based filtering, discarding loss gradients from positions where the teacher's top-1 probability falls below a threshold. AdaSwitch (Peng et al., 2025) dynamically switches between exploration (student generates freely) and guidance (teacher provides dense supervision) based on the student's cumulative error on the current prefix, balancing the exploration benefit of on-policy generation against the stability of teacher guidance. Token-Adaptive KD (Xie et al., 2025) computes per-token importance weights using the gradient magnitude of the KL loss, concentrating compute on tokens where the teacher-student gap is largest.

Token-Adaptive KD. Xie et al. (2025) compute per-token importance weights using the gradient magnitude of the KL loss, concentrating compute on tokens where the teacher-student gap is largest. This approach is complementary to entropy-based routing (ToDi, Entropy-Aware): while the latter selects *which divergence* to use at each position, Token-Adaptive KD selects *how much gradient signal* to propagate, providing a second dimension of per-token adaptation that can be combined with adaptive divergence selection.

Pre-training distillation. MiniPLM (Gu et al., 2025) extends on-policy distillation to the *pre-training* stage, showing that distilling during pre-training (rather than post-training) yields stronger foundations for downstream fine-tuning. This represents a qualitative shift: rather than being a post-hoc compression step, OPD becomes an integral part of the model creation pipeline. Gemma 2 (Gemma Team et al., 2024) validates this approach at industrial scale, embedding KD directly into the pre-training loop for its 27B→9B→2B model cascade.

Cross-architecture alignment. A critical challenge arises when teacher and student use different tokenizers or vocabulary sizes. Direct KL divergence over mismatched vocabularies is undefined. DSKD (Zhang et al., 2025b) solves this through *dual-space* knowledge distillation, introducing projectors that map between teacher and student representation spaces:

$$\mathcal{L}_{\text{DSKD}} = D_{\text{KL}}(P_{T\to S} \parallel P_S) + D_{\text{KL}}(P_{S\to T} \parallel P_T) \tag{15}$$

where $P_{T\to S}$ denotes the teacher's hidden states projected into the student's space and decoded through the student's prediction head. An exact token alignment algorithm handles vocabulary mismatches by identifying identical tokens across tokenizers. The key insight is that vocabulary tokens with similar semantic embeddings should receive similar probability mass, even when the tokenization schemes differ. This enables distillation across fundamentally different architectures (e.g., Llama teacher to Qwen student) that share no vocabulary overlap.

Cross-Tokenizer KD (Boizard et al., 2025) takes a complementary approach based on optimal transport, computing an alignment matrix between teacher and student token sequences that minimizes the earth mover's distance in latent space:

$$\mathcal{L}_{\text{CrossTok}} = \inf_{\pi\in\Pi(P_S,P_T)} \mathbb{E}_{(z_S,z_T)\sim\pi}\left[\|W_{S\to T}z_S - z_T\|_2^2\right] \tag{16}$$

The distinction between hidden-state alignment (DSKD) and vocabulary-space alignment (Cross-Tokenizer KD) highlights a fundamental design trade-off. Hidden-state alignment preserves richer representational structure but requires maintaining two projectors during training. Vocabulary-space alignment is more lightweight but operates at a level where information has already been compressed through the vocabulary bottleneck. In either case, the alignment maps must be *co-trained* with the student, because the student's representation space shifts during on-policy training as its policy evolves, a form of representation drift absent in off-policy settings. Concrete Score Matching (Kim et al., 2026b) provides yet another solution, distilling via a continuous score function that bypasses the discrete vocabulary mismatch entirely. Together, these methods ensure that OPD is architecture-agnostic, not limited to same-family distillation.

Signal isolation. Not all information in the teacher's logits is equally useful. Delta-KD (Ko et al., 2026) introduces the principle of *signal isolation*: rather than distilling the teacher's full distribution, it distills only the *differential shift* between a base model and its instruction-tuned variant. The base model's distribution captures generic language modeling knowledge that the student already possesses (or should acquire through standard pre-training), so distilling it is redundant at best and harmful at worst. By extracting only the instruction-following delta, Delta-KD provides a cleaner supervisory signal that transfers the teacher's alignment without overwriting the student's foundation.

Stable target reformulation. A subtle but important challenge in white-box OPD is that the teacher's logit distribution can be *overconfident* on tokens outside the student's reachable space, creating optimization targets that are simultaneously precise and useless. Veto (Jang et al., 2026) (Adaptive Target Reformulation) addresses this by constructing a geometric bridge in logit space: a tunable parameter α serves dual roles as an Adaptive Gradient Veto (suppressing pathological gradients on low-confidence tokens, addressing Forward KL instability) and a Decisiveness Knob (balancing reward-driven performance with output diversity, addressing Reverse KL mode collapse). By dynamically reformulating the teacher's target distribution based on the student's current capability, it smooths the teacher's predictions at positions where the student cannot yet match them. This prevents the pathological gradient behavior where an overconfident teacher creates extremely sharp loss landscapes that the student oscillates around without converging. On reasoning and generation tasks, Veto consistently outperforms GKD and DistiLLM.

Selective supervision and prompt-based distillation. PromptKD (Kim et al., 2024) takes a radically different approach to white-box distillation: rather than matching the teacher's output distribution directly, it prepends learnable prompt tokens (adding only 0.0007% of the teacher's parameters) that cause the teacher to produce "student-friendly" distributions more accessible to the student's limited capacity. This implicitly addresses the capacity gap problem: when the teacher's distribution is too complex for the student to match, the standard KL objective drives the student toward a blurred average of the teacher's modes. By simplifying the teacher's output before distillation, PromptKD reduces this gap at its source.

Temporally adaptive targets. TAID (Shing et al., 2025) constructs a time-dependent intermediate distribution $P_t = (1 - \lambda_t)P_\theta + \lambda_t P_T$ where λ_t gradually increases during training, effectively starting with a target close to the student (easy to match) and progressively approaching the full teacher distribution. This temporal curriculum over the *target itself* (rather than over the input prompts as in PACED) provides a complementary axis of control: PACED asks "which problems to train on," while TAID asks "how much of the teacher's knowledge to reveal at each stage." The combination of prompt-level curriculum (PACED) with target-level curriculum (TAID) represents the current frontier of training dynamics optimization.

The progression from PACED (problem-level curriculum) through AdaSwitch (token-level timing) to TAID (target-level interpolation) and PromptKD (teacher-level adaptation) traces an important design spectrum: curriculum can operate at four levels, all of which address the same underlying bottleneck of keeping the difficulty gap between teacher signal and student capability in the productive learning range.

5.2 Black-Box and API-Constrained Distillation

When only API access is available, the student observes the teacher's top-1 text output (or at best, top-k log-probabilities) rather than the full distribution. This information bottleneck eliminates token-level divergence matching and forces methods to operate at the sequence level. Despite this limitation, black-box methods have proven surprisingly effective, suggesting that much of the teacher's knowledge is recoverable from text-only observations.

Distilling Step-by-Step. Hsieh et al. (2023) demonstrated an influential early approach: extracting not just the teacher's final answers but its intermediate reasoning steps (rationales) as additional supervision. The student is trained with a multi-task objective combining answer prediction and rationale generation, enabling it to outperform the teacher on downstream tasks with significantly less data. This work established the principle that *how* the teacher reasons is often more valuable than *what* it concludes, a principle that pervades modern OPD.

Distilling Step-by-Step. Hsieh et al. (2023) demonstrated an influential early approach: extracting not just the teacher's final answers but its intermediate reasoning steps (rationales) as additional supervision. The student is trained with a multi-task objective combining answer prediction and rationale generation, enabling it to outperform the teacher on downstream tasks with 1,000× less data. This work established the principle that *how* the teacher reasons is often more valuable than *what* it concludes.

Adversarial distribution matching. GAD (Ye et al., 2025) reformulates the distillation problem as a generative adversarial game:

$$\max_{G} \min_{D} \; V(G, D) = \mathbb{E}_{(x, y^T) \sim \mathcal{T}} \left[-\log \sigma \Big(D(y^T) - D(G(x)) \Big) \right] \tag{17}$$

A discriminator D (initialized from the student with a scalar prediction head) distinguishes student rollouts from teacher API outputs via a Bradley-Terry preference model, while the generator G maximizes V via REINFORCE using the discriminator score as an evolving on-policy reward signal. Unlike white-box methods that match full distributions, GAD extracts knowledge through a scalar quality signal, trading distributional fidelity for API compatibility. It inherits the training instability of adversarial objectives, as the discriminator can overfit to stylistic cues rather than semantic quality, but achieves consistent gains over SFT baselines.

Verbal feedback and curriculum. Lion (Jiang et al., 2023) exploits the teacher's capacity as a natural language critic through a three-stage adversarial loop: (1) *imitation*, where the student is fine-tuned on teacher-generated responses, (2) *discrimination*, where the teacher identifies instructions on which the student still falls short, and (3) *generation*, where the teacher creates harder instructions targeting the student's identified weaknesses. This closed-loop curriculum progressively narrows the teacher-student gap without requiring any gradient information from the teacher, making it applicable to any instruction-following API. Lion-13B achieves competitive performance on BIG-Bench Hard and AGIEval using only 70K training examples.

Didactic-to-constructive learning. DAIL (Mendes et al., 2026) bridges the gap between high-quality expert solutions that are out-of-distribution for the student and the student's actual reasoning space. The core challenge is that simply fine-tuning on expert solutions yields poor transfer because the student never learns to navigate its own reasoning space. DAIL applies a two-step process: first transforming expert solutions into detailed, in-distribution reasoning traces through inverse reinforcement learning (inferring the implicit reward function guiding the teacher's generation choices from observed text trajectories), then applying a contrastive objective to focus learning on expert methodologies rather than surface patterns. Using fewer than 1,000 expert solutions, DAIL achieves 10–25% pass@k gains. This result underscores a principle that recurs throughout this survey: the quality of the supervision signal matters less than its *alignment* with the student's current distribution.

Distribution-Aligned Sequence Distillation (DASD) (Yan et al., 2026) critically reexamines the dominant practice of SFT on teacher-generated reasoning traces, identifying three fundamental limitations: (1) inadequate representation of the teacher's sequence-level distribution from single best-of-n samples, (2) misalignment between teacher output distribution and student capacity, and (3) exposure bias from teacher-forced training. Its enhanced distribution-aligned pipeline addresses all three by generating multiple diverse teacher traces per prompt and aligning the student's generation distribution with the teacher's through sequence-level matching, achieving state-of-the-art performance using only 448K training samples. DDT (Zhang et al., 2026c) complements DASD's empirical findings with a theoretical framework explaining why RL generalizes better than SFT for reasoning tasks:

the key factor is RL's on-policy data generation, which keeps training aligned with the model's evolving distribution rather than a static snapshot.

Verbal score distillation. OVD (Xiong et al., 2026) replaces token-level probability matching entirely with trajectory matching using discrete verbal scores (0–9) from teacher models. This eliminates the requirement for token-level alignment between vocabularies, dramatically reduces memory consumption by avoiding logit storage, and delivers up to +12.9% absolute EM improvement on web QA and +25.7% on math benchmarks with a single rollout sample. The surprising effectiveness of such coarse supervision demonstrates that on-policy exploration itself provides a substantial fraction of the learning signal, with the teacher's role being more to *select among student trajectories* than to *correct individual tokens*.

Preference-based formulations. ORPO-Distill (Singh et al., 2025) maps black-box distillation onto preference optimization, generating candidate responses from the student and querying the teacher API for pairwise rankings. The mixed-policy strategy, combining on-policy student traces with off-policy teacher traces, outperforms both pure off-policy and pure on-policy alternatives, suggesting that the contrastive structure of preference learning (learning what *not* to do alongside what to do) provides complementary signal to the positive-only feedback of score-based methods. ThinkTuning (RRV et al., 2025) occupies an intermediate position between black-box distillation and RL: rather than providing scores after the fact, a same-size teacher reviews the student's incorrect answers *during* the GRPO training loop and provides short corrective hints (rather than complete solutions). This interleaving of corrective feedback episodes with RL training acts as an implicit privileged signal that shapes policy gradient updates, and is particularly effective for base models that lack strong prior reasoning behaviors, achieving +3.85% over zero-shot baselines across benchmarks.

Off-policy knowledge integration. A related challenge is incorporating knowledge from off-policy sources (e.g., pre-existing teacher-generated datasets) into an on-policy framework. LUFFY (Yan et al., 2025) addresses this by extending GRPO to a mixed-policy objective that incorporates off-policy reasoning traces alongside the student's on-policy rollouts. The critical design challenge is preventing *superficial imitation*: without correction, the student tends to copy high-probability tokens from off-policy traces while ignoring low-probability but critical reasoning steps. LUFFY's policy shaping via regularized importance sampling reweights gradients to emphasize learning from unfamiliar yet effective actions, achieving +6.4 average points over standard RLVR.

5.3 Self-Distillation: Privileged Information

Self-distillation eliminates the need for an external teacher entirely. Instead, the model exploits asymmetries *within itself* to construct a training signal. The first family of self-distillation methods creates this asymmetry through *privileged information* (PI): training-time access to context that is unavailable at inference. The central insight is that conditioning the same model on additional context produces a strictly stronger policy that can supervise the unconditioned version, turning a single network into both teacher and student.

Ground-truth as PI. OPSD (Zhao et al., 2026b) constructs the teacher policy by conditioning on both the problem x and the ground-truth answer $y^\star$, while the student conditions only on x. The student generates on-policy rollouts, and the teacher provides dense token-level supervision:

$$\mathcal{L}_{\text{OPSD}}(\theta) = \mathbb{E}_{(x,y^\star)\sim\mathcal{S}}\mathbb{E}_{\hat{y}\sim p_\theta(\cdot|x)}\left[\sum_{n=1}^{|\hat{y}|} D\big(p_\theta(\cdot|x,y^\star,\hat{y}_{<n}) \parallel p_\theta(\cdot|x,\hat{y}_{<n})\big)\right] \tag{18}$$

On competition-level math (AIME 2024/2025), OPSD matches or exceeds GRPO at 4B and 8B scales while using only a single rollout per problem versus GRPO's eight, though at 1.7B the gains are minimal, indicating that sufficient model capacity is necessary for the PI mechanism to provide useful supervision.

HDPO (Ding, 2026) addresses a specific failure mode of OPSD: when all student rollouts fail (the "cliff prompt" problem), the policy gradient vanishes entirely. HDPO augments GRPO with a JSD-based self-distillation term on ground-truth-conditioned rollouts, ensuring non-zero gradients even at the boundary of the student's competence.

Relaxing the ground-truth requirement. Ground-truth labels are not always available. GATES (Stein et al., 2026) demonstrates that *any* training-time context can serve as PI by having a single model act as tutor (conditioned on a source document) and student (answering from the question alone). Because source documents do not guarantee correct answers, GATES introduces a consensus-based gating mechanism that suppresses the gradient when tutor uncertainty is high, preventing the model from reinforcing unreliable self-supervision. Penaloza et al. (2026) generalize the OPSD and GATES paradigms into a unified theoretical framework, formalizing the conditions under which privileged self-distillation provably improves over standard training.

Behavioral PI: context distillation. The PI paradigm extends beyond factual ground truth to behavioral instructions. OPCD (Ye et al., 2026b) treats system prompts, exemplars, and retrieval context as privileged information, enabling a model to internalize behaviors that would otherwise require expensive in-context demonstrations at inference time. OPSDL (Anonymous, 2026a) extends this to long-context settings: applying on-policy self-distillation to enhance long-context capabilities without relying on high-quality supervision data or sparse sequence-level rewards, enabling stable optimization for context length scaling beyond what standard post-training methods achieve. OEL (Ye et al., 2026a) extends to continuous post-deployment learning, using interaction outcomes as dynamic PI that updates the model's teacher policy as it accumulates experience, closing a loop that OPCD opened.

Reasoning compression via PI. CRISP (Sang et al., 2026) demonstrates that a model prompted to "be concise" can serve as a teacher for its own verbose version, reducing chain-of-thought token count by 57–59% on MATH-500 while *improving* accuracy by 9–16 percentage points. This reveals that much of the verbosity in distilled reasoning models is a trainable inefficiency, not a necessary feature of competent reasoning.

5.4 Self-Distillation: Self-Play

A more radical question is whether a model can improve using nothing but its own rollouts, with no PI, no external feedback, and no additional context. The answer is a qualified yes, subject to important saturation limitations.

Self-play as a two-player game. SPIN (Chen et al., 2024) casts self-improvement as an adversarial game between successive checkpoints, training the updated model to distinguish its previous iteration's generations from human-written responses:

$$\mathcal{L}_{\text{SPIN}} = \mathbb{E}_{x, y \sim p_{\text{data}}, y' \sim p_{\theta_t}} \left[\ell \left(\lambda \left(\log \frac{p_{\theta_{t+1}}(y|x)}{p_{\theta_t}(y|x)} - \log \frac{p_{\theta_{t+1}}(y'|x)}{p_{\theta_t}(y'|x)} \right) \right) \right] \tag{19}$$

SPIN provides a theoretical convergence guarantee: the global optimum of $\mathcal{L}_{\text{SPIN}}$ is achieved if and only if $p_{\theta_{t+1}} = p_{\text{data}}$, i.e., the student's distribution exactly matches the reference distribution. Crucially, at this fixed point, the loss is zero and the gradient vanishes, so the iterative process is *stable*: once the student reaches the target distribution, further iterations cannot destabilize it. The game-theoretic interpretation is that p_{data} is the unique Nash equilibrium of the two-player game, and SPIN's iterative updates correspond to fictitious play converging to this equilibrium. Starting from Zephyr-7B-SFT, it improves MT-Bench from 5.94 to 6.78 over 3 iterations, with diminishing returns at each round. However, this convergence guarantee comes with a fundamental limitation: SPIN converges to p_{data}, not to the optimal policy. If the reference data is sub-optimal (as human-written responses often are), SPIN will faithfully reproduce those sub-optimalities, revealing that pure self-play recycles existing knowledge rather than generating new knowledge.

IRIS (Anonymous, 2026b) provides a unifying framework for this family by replacing the implicit KL divergence in SPIN with an interpolative Rényi divergence parameterized by $\alpha \in (0, \infty)$, recovering SPIN ($\alpha \to 1$, KL regime), SPACE ($\alpha \to 0.5$, Jensen-Shannon regime), and SPIF ($\alpha \to 2$, χ^2-regularized regime) as special cases. The Rényi parameter controls the exploration-exploitation tradeoff: lower α encourages broader distributional coverage (exploration) while higher α concentrates updates on the most likely improvements (exploitation). This unification reveals that the various self-play methods in the literature differ only in their implicit divergence choice, not in their fundamental mechanism, and suggests that α-tuning can be used as a meta-parameter for adapting self-play behavior to task requirements.

This motivates the external grounding mechanisms of Section 5.5.

Temporal self-supervision. SDFT (Shenfeld et al., 2026) positions on-policy self-distillation as a mechanism for forgetting resistance via implicit KL regularization against the model's own prior checkpoint, converting the model's optimization trajectory into a regularization resource for continual learning.

Self-play for efficiency. MTP via self-distillation (Kirchenbauer et al., 2026) converts a pre-trained autoregressive model into a multi-token predictor by training an auxiliary prediction head on the model's own next-token predictions, achieving $> 3\times$ faster decoding at $< 5\%$ accuracy drop. This demonstrates that self-distillation can target *architectural* improvements (inference speed) rather than *behavioral* improvements (task accuracy), expanding the scope of what OPD can optimize.

On-Policy SFT (Zhao et al., 2026a) provides a surprising simplification: by filtering self-generated responses for correctness (via a verifier) and conciseness (via length truncation at the median successful length), standard SFT on this filtered on-policy data matches or exceeds full RL training. The key insight is that the combination of on-policy generation (which naturally produces diverse solution strategies) with outcome filtering (which selects the best strategies) implicitly performs a form of evolutionary optimization that requires no gradient through the reward function. This "generate-filter-train" paradigm dramatically reduces implementation complexity compared to full RL-based distillation while retaining the core benefit of on-policy data: distributional alignment.

Self-Distilled RLVR (Yang et al., 2026a) combines self-distillation with verification-based RL, showing that the two mechanisms are complementary: self-distillation provides the dense signal for stable training while RLVR provides the sparse but accurate outcome signal for quality improvement.

Minimalist self-distillation. SSD (Zhang et al., 2026d) requires no RL, no verifier, no external teacher, and no code execution. It samples solutions at training-time temperature, fine-tunes via standard SFT, and reshapes token distributions in a context-dependent manner. On LiveCodeBench v6, SSD improves Qwen3-30B-Instruct from 42.4% to 55.3%, revealing that when the base model is sufficiently capable, even the simplest self-play yields substantial gains. The critical requirement is not a sophisticated algorithm but sufficient *diversity* in the model's sampling distribution to produce high-quality variants that can serve as training targets.

However, this conclusion also reveals the fundamental boundary of pure self-play: when diversity is achieved through temperature sampling alone, the model's improvement is ultimately bounded by the information already present in its pre-training distribution. Bridging this boundary requires either connecting the model to external reality (Section 5.5) or internally generating privileged information through the self-play process itself (π-Play below).

Unifying PI and self-play. π-Play (Zhang et al., 2026e) unifies privileged information and self-play within a single multi-agent self-evolution framework. The key observation is that self-play naturally produces a byproduct: during task generation, the examiner agent constructs a question construction path (QCP) that provides high-quality PI at no additional cost. Three co-evolving agents (examiner π_ϕ^E, teacher π_ψ^T conditioned on QCP, student π_θ^S without it) are jointly optimized through alternating updates. As the student strengthens, the examiner generates harder tasks, and the teacher's behavior is softly constrained to track the student via a KL penalty, establishing a self-adjusting curriculum that requires no external data. On Qwen3 (4B, 8B), data-free π-Play surpasses supervised search agents (Search-R1 by 5–15%) and improves efficiency by 2–3$\times$ over conventional self-play (Dr.Zero), with the largest gains on multi-hop benchmarks where the teacher's token-level credit assignment provides the most leverage. CoPD (Zhang et al., 2026f) extends this multi-agent paradigm to collaborative settings where multiple student agents jointly distill from each other's on-policy rollouts, enabling knowledge sharing across different model specializations without requiring a single dominant teacher.

π-Play demonstrates that the self-play saturation problem can be mitigated *without* external feedback by internally generating privileged information through the self-play process itself. This mechanism is orthogonal to the external-verification strategies of Section 5.5, suggesting that the two approaches could be combined for even stronger self-improvement.

5.5 Self-Distillation: External Feedback

Pure self-play is inherently bounded by the information already present in the model's pre-training distribution. External signals, such as compiler outputs, test results, and textual critiques, ground self-distillation in objective reality and bridge the gap to reinforcement learning.

From binary rewards to dense supervision. SD-ZERO (He et al., 2026) converts sparse verifier signals into dense self-supervision through self-revision: after generating a solution and receiving a binary reward, the model generates a *revised* solution conditioned on the original attempt. When the revision succeeds where the original failed, the (original, revision) pair provides a natural preference signal without requiring any external teacher. SDPO (Hübotter et al., 2026) extends this beyond binary feedback to *structured textual* feedback: runtime errors, failing unit tests, and LLM judge evaluations construct fine-grained credit assignment that attributes success or failure to specific reasoning steps rather than entire trajectories. RL from Text Feedback (RLTF) (Song et al., 2026) pushes further with free-form natural language critiques from an automated judge, training the model's single-turn policy to match its own feedback-conditioned second-turn generations.

The progression from SD-ZERO's binary signal through SDPO's structured errors to RLTF's free-form critiques traces a clear trajectory: as external feedback becomes richer, the self-distillation objective becomes more precisely targeted, enabling the model to correct specific failure modes rather than globally adjusting its policy. Sample-Routed Policy Optimization (SRPO) (Li et al., 2026b) demonstrates that these approaches can be composed by routing: correct samples follow GRPO's reinforcement path (reward-weighted likelihood maximization) while failed samples follow SDPO's targeted logit-level correction. The insight is that correct and incorrect samples carry distinct types of learning signal, and treating both identically wastes the distinctive information each carries. SRPO raises the average by 3.4% over GRPO and 6.3% over SDPO alone, confirming that the optimal external-feedback strategy is a routing function matching each sample to its most informative objective.

Environmental grounding. RLTF and related methods use compiler outputs, unit test results, and symbolic math verifiers as external feedback that shatters self-reinforcing echo chambers. Even if the model becomes highly confident in a flawed derivation, the verifier assigns $R = 0$, providing an unambiguous correction signal that pure self-play cannot generate.

Saturation analysis. Kim et al. (2026a) trace self-distillation degradation in mathematical reasoning to the suppression of *epistemic verbalization*: the model's expression of uncertainty during reasoning. When self-distillation shortens reasoning traces, it disproportionately removes hedging phrases and uncertainty markers that serve as implicit "attention flags" for difficult sub-problems. The result is shorter but less calibrated reasoning, where the model confidently commits to flawed steps. This connects to the teacher uncertainty challenge in Section 7: an overconfident student produces unreliable self-play targets, and the saturation problem ultimately requires either external grounding or the internally generated PI mechanism of π-Play to overcome.

The progression across signal sources, from white-box logits through black-box APIs to self-distillation, mirrors a fundamental trade-off between signal density and autonomy. White-box methods provide the richest supervision ($|V|$-dimensional distributions at every token) but require same-organization deployment and co-location of teacher and student. Black-box methods sacrifice distributional information but operate across organizational boundaries, with recent methods (GAD, OVD) recovering surprisingly effective proxy signals. Self-distillation eliminates external dependencies entirely but is bounded by the model's pre-existing capabilities unless augmented by external verification or internally generated PI. This continuum is not merely descriptive: it predicts where future methods will have the most impact. The white-box extreme is well-explored (GKD, DistiLLM, DSKD are mature), while the self-distillation extreme still lacks principled convergence guarantees for complex reasoning. The greatest innovation potential lies at the boundaries: methods like ThinkTuning that blur the line between black-box feedback and RL, or methods like π-Play that blur the line between self-play and externally grounded verification.

6 Training Dynamics and Efficiency

Given an objective function (Section 4) and a signal source (Section 5), the final engineering decision is *how to stabilize and accelerate the on-policy training loop*. On-policy generation introduces unique optimization challenges absent from standard supervised fine-tuning: non-stationary data distributions (the student's policy shifts during training, making older rollouts stale), gradient signal-to-noise ratio (SNR) collapse on hard prompts (where all rollouts fail and no useful gradient emerges), and the computational overhead of autoregressive student rollouts followed by teacher scoring (typically 4–5$\times$ more expensive than off-policy training).

These challenges are not independent. SNR collapse on hard prompts wastes compute, motivating curriculum strategies that avoid such prompts. Non-stationarity invalidates cached teacher signals, motivating efficient online scoring. And the compute overhead of autoregressive generation creates a systems-level bottleneck that requires architectural solutions. The methods in this section address these challenges through three complementary mechanisms.

6.1 Token and Sample Weighting

Not all teacher supervision is equally reliable in the on-policy setting. The core problem is the *flawed prefix trap*: when a student generates an erroneous token early in a sequence, all subsequent teacher predictions are conditioned on this out-of-distribution prefix. Because the teacher has never been trained on such prefixes, its conditional distribution $p_T(\cdot|x, \hat{y}^{\text{bad}}_{<t})$ becomes noisy and unreliable. Naively matching this corrupted signal propagates errors rather than correcting them.

Teacher reliability filtering. TIP (Xu et al., 2026b) (Token Importance Profiling) provides a principled framework for token weighting by organizing importance along two axes: student entropy h_t and teacher-student divergence δ_t. Crossing these axes yields four quadrants with qualitatively distinct learning roles: high-entropy, high-divergence tokens (Q1) carry the dominant corrective signal, high-entropy, low-divergence tokens (Q2) indicate agreement on uncertainty, low-entropy, high-divergence tokens (Q3) represent *overconfident errors* where the student is certain but wrong, and low-entropy, low-divergence tokens (Q4) are genuinely mastered. The key theoretical result shows that Q3 tokens are structurally invisible to any entropy-only weighting scheme, because no non-decreasing function of entropy with $f(0) = 0$ can distinguish them from Q4. TIP's parameter-free Soft-OR score $s_t = \hat{h}_t + \hat{\delta}_t - \hat{h}_t \cdot \hat{\delta}_t$ recovers Q3 coverage while preserving Q1/Q2 sensitivity, achieving consistent improvements at 50% token retention across three model families with capacity gaps from 2$\times$ to 9$\times$.

SCOPE (Zheng et al., 2026) elevates this analysis from tokens to *rollouts*, revealing the "flawed prefix trap": when the student produces an erroneous prefix, the teacher's conditional distribution becomes poorly calibrated because it has never been trained on such inputs. SCOPE's dual-path architecture routes rollouts by correctness: incorrect trajectories receive teacher-perplexity-weighted KL distillation (down-weighting unreliable guidance), while correct trajectories receive student-perplexity-weighted MLE (concentrating reinforcement at the capability boundary). This addresses diversity collapse, the counterintuitive phenomenon where Pass@1 improves at the expense of Pass@k, yielding +5.5% over standard OPD.

SelecTKD (Huang et al., 2025) applies confidence-based filtering at the token level, discarding loss gradients from positions where the teacher's top-1 probability falls below a threshold. The reasoning is that low teacher confidence indicates either genuine ambiguity (where any student token is acceptable) or distributional confusion (where the teacher cannot provide useful supervision), and in neither case does the gradient carry reliable information.

AdaSwitch (Peng et al., 2025) introduces a dynamic switching mechanism between pure exploration and guided distillation. Unlike static filtering that simply attenuates unreliable signals, AdaSwitch maintains a running estimate of the student's cumulative prefix quality and switches between on-policy exploration (when the prefix is reasonable) and off-policy guidance (when the prefix has drifted too far). This binary switching is computationally simpler than continuous weighting while empirically achieving similar stability.

Sample-level weighting. Beyond token-level filtering, several methods weight entire samples based on their informativeness. The flawed prefix trap manifests most severely on *hard* prompts where the student's pass rate approaches zero. On these prompts, nearly all rollouts contain early catastrophic errors, and the resulting teacher signals are dominated by noise. Methods that uniformly sample prompts waste most of their gradient budget on uninformative hard prompts or trivially correct easy prompts, with the productive learning concentrated on prompts at the boundary of the student's competence.

Failure-mode analysis. Fu et al. (2026) provide a systematic empirical analysis of why sampled-token OPD is fragile in long-horizon settings. They identify three failure modes: (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on student-generated prefixes that deviate from the teacher's training distribution, and (3) distortions caused by tokenizer or special-token mismatch. Their proposed fix, teacher top-K local support matching via truncated Reverse KL with top-p rollout sampling and special-token masking, yields more stable optimization across math reasoning and agentic training.

The token and sample weighting methods above operate *reactively*: they attenuate the loss signal *after* the student has already generated a potentially uninformative rollout. A more proactive approach asks: can we select prompts *before* generation to maximize the expected information gain per rollout? This motivates the curriculum methods below.

6.2 Curriculum and Difficulty Adaptation

The sample weighting problem motivates a more principled solution: *curriculum design* that actively selects prompts matching the student's current competence level.

Competence-boundary sampling. PACED (Xu et al., 2026a) models prompt difficulty dynamically using a Beta-kernel distribution centered on the student's current pass rate. The curriculum progressively shifts toward harder prompts as the student improves, but critically avoids prompts where the pass rate is near zero (where gradient SNR vanishes) or near one (where the student has already mastered the content). This "frontier sampling" strategy concentrates the gradient budget on the narrow band of prompts where learning is most efficient.

The mathematical justification connects to the gradient SNR analysis: for a prompt with pass rate p, the expected gradient magnitude scales as $p(1-p)$, which is maximized at $p = 0.5$ and vanishes at both extremes. PACED's Beta-kernel sampling directly optimizes for this quantity, providing a theoretically grounded alternative to uniform prompt sampling. Concretely, PACED maintains a running estimate of per-prompt difficulty $\hat{p}_i$ (computed from the last m rollouts) and samples each prompt with probability proportional to $\text{Beta}(\hat{p}_i; \alpha, \beta)$ where (α, β) are hyperparameters controlling the width of the competence frontier. Early in training, wide kernels cast a broad net. As training converges, narrow kernels focus on the diminishing frontier of unsolved problems. This reduces wasteful rollouts on already-mastered problems by 40–60% compared to uniform sampling, translating directly into compute savings proportional to the curriculum efficiency.

Self-adjusting curricula. π-Play's multi-agent framework (Section 5.4) provides an alternative curriculum mechanism: as the student strengthens, the examiner agent automatically generates harder tasks, creating a self-adjusting difficulty ladder that requires no external difficulty estimation. PRISM (Li et al., 2026d) provides a more structured approach: progressive reasoning via iterative self-distillation with a mastery-based curriculum that only advances the student to harder problems once it demonstrates consistent performance on the current difficulty tier. PAINT (Wang et al., 2026c) addresses the prefix-alignment problem in instruction tuning, ensuring that the student's on-policy rollouts are conditioned on properly formatted instruction prefixes that match the deployment format, preventing a subtle form of distribution mismatch between training and inference.

Off-policy cold start. Li et al. (2026e) identify that OPD can fail catastrophically when the student's initial policy is too far from the teacher's distribution (the "thinking-pattern mismatch" problem). Their proposed solution is an off-policy cold-start phase: a brief period of standard SFT on teacher-generated data that reduces the pattern gap before switching to on-policy training. This two-phase approach, off-policy warmup followed by on-policy refinement, has become the standard industrial pattern (Section 8).

Retaining knowledge during on-policy training. A complementary concern is that on-policy training can cause *catastrophic forgetting* of capabilities the student acquired during pre-training. Chen et al. (2025) ("Retaining by Doing") demonstrate that on-policy data generation itself serves as an implicit rehearsal mechanism: by generating sequences from its own distribution, the student naturally revisits and reinforces its existing capabilities while learning new ones. This insight suggests that the on-policy generation step provides a dual benefit: it both exposes the student to its own error states (enabling correction) and rehearses its existing capabilities (preventing forgetting), offering a theoretical explanation for why on-policy methods empirically suffer less from catastrophic forgetting than sequential fine-tuning approaches.

Semantic bootstrapping. Semantic Soft Bootstrapping (Mitra & Ulukus, 2025) provides a different perspective on curriculum design: rather than adapting the *difficulty* of prompts, it adapts the *semantic granularity* of supervision. The method progressively refines the teacher's supervision from coarse semantic alignment (matching high-level reasoning structure) to fine-grained token-level matching, allowing the student to first acquire the overall reasoning pattern before optimizing the precise token distributions.

The hybrid SFT+OPD pipeline. The off-policy cold start (Li et al. (2026e)) combined with on-policy refinement has become the de facto industrial recipe. Empirically, the optimal transition point from off-policy to on-policy training occurs when the student's generation distribution achieves sufficient overlap with the teacher's (typically measured by a token-level agreement rate >60%). Transitioning too early wastes on-policy compute on uninformative rollouts (the student is too weak for meaningful self-generation). Transitioning too late leaves the student stuck in the off-policy ceiling. Qwen3 (Yang et al., 2025a) uses approximately 20% off-policy warmup followed by 80% on-policy refinement. DeepSeek-R1 (DeepSeek-AI et al., 2025) uses 100% off-policy (no transition), which works at extreme teacher capability but leaves performance on the table for reasoning tasks where the student could exceed the teacher through on-policy exploration.

6.3 Compute Optimization

OPD requires generating student rollouts *and* scoring them with the teacher at every training step, creating a compute overhead of 3–8$\times$ compared to off-policy SFT. Several methods target this bottleneck.

The cost breakdown for a typical white-box OPD step consists of:

1. **Student rollout** ($\sim$40% of step time): autoregressive generation of $B \times K$ completions (batch size B, K rollouts per prompt), memory-bound due to KV cache growth.
2. **Teacher scoring** ($\sim$35% of step time): forward pass through the teacher to obtain full-vocabulary logits on student-generated sequences, compute-bound.

3. **Student update** ($\sim$25% of step time): standard backward pass computing gradients of the distillation loss, dominated by optimizer state memory.

Each component admits independent optimization, and the methods below attack different bottlenecks.

Prefix truncation. Fast OPD (Zhang et al., 2026a) observes that exposure bias is concentrated in the early tokens of a sequence, where errors compound most severely. It truncates on-policy rollouts to a prefix of length k and reverts to off-policy KD for the remainder, achieving comparable quality at substantially reduced cost. The optimal truncation point k depends on the task: for mathematical reasoning, where errors cascade rapidly, short prefixes suffice. For open-ended generation, longer on-policy segments are needed.

Offline teacher caching. Lightning OPD (Wu et al., 2026) decouples student optimization from teacher inference entirely by precomputing teacher distributions into an offline memory bank and asynchronously refreshing them. When teacher distributions are *consistent* (i.e., the teacher's output depends primarily on the prompt and only weakly on the specific student prefix), the cached distributions remain valid even as the student's policy evolves. Lightning OPD demonstrates that offline distillation under teacher consistency can match online OPD at 4$\times$ lower cost, significantly lowering the barrier for academic research.

Speculative knowledge distillation. Speculative KD (Xu et al., 2025c) adapts speculative decoding to the distillation setting: the student generates candidate tokens that the teacher *verifies and selectively replaces*, amortizing the teacher's inference cost across multiple student tokens per teacher forward pass. DistillSpec (Zhou et al., 2023) inverts this: instead of using speculation to speed up training, it uses on-policy KD to improve the draft model for speculative decoding at *inference* time. The expected speedup:

$$E[S] = \frac{\mathbb{E}[K_{\text{accepted}}] + 1}{1 + c} \quad \text{where } c = \frac{\text{cost(student forward)}}{\text{cost(teacher forward)}} \tag{20}$$

This reveals a virtuous cycle: better distillation produces better draft models, which enable faster speculative decoding, which in turn enables more efficient distillation, progressively reducing the computational barrier.

GPU memory management. Beyond FLOPs, the most severe constraint for white-box OPD is GPU memory. Holding a 70B teacher alongside a 7B student requires the teacher's weights ($\sim$140 GB in BF16), the student's weights and optimizer states ($\sim$84 GB), and the full-vocabulary logits tensor $[B, T, |V|]$ for distillation. Peak memory can easily exceed an 8×80 GB H100 node. Practitioners alleviate this via: (1) *teacher quantization*, serving the teacher in FP8 or INT4 for a 2–4$\times$ memory reduction at negligible precision loss, (2) *logit offloading*, immediately streaming logits to CPU RAM or retaining only the top-k entries (since 99% of the probability mass is typically concentrated in fewer than 100 tokens), and (3) *aggressive gradient checkpointing* to minimize student activation memory during the backward pass. Combining these techniques is essential for viable white-box OPD on standard clusters.

Concrete cost example. To ground these formulas, consider distilling a 70B teacher into a 7B student on 8$\times$H100 GPUs. Off-policy over 1B tokens: the teacher generates the dataset offline ($\sim$200 GPU-hours), the student trains for $\sim$100 GPU-hours, totaling $\sim$300 GPU-hours. On-policy over 1B tokens: each step requires (1) student generation ($\sim$3$\times$ forward-pass cost due to autoregressive decoding), (2) teacher scoring (one 70B forward pass), and (3) student backward pass, yielding $\sim$1,200–1,500 GPU-hours, a 4–5$\times$ overhead. However, for reasoning tasks, on-policy methods consistently outperform off-policy SFT, and the off-policy performance ceiling is strictly lower because static datasets cannot cover the combinatorial space of valid reasoning paths. This "ceiling gap" widens with task complexity, providing the strongest cost-benefit justification for on-policy methods.

7 Understanding OPD: Theory, Failure Modes, and Cost

The preceding sections addressed the *how* of on-policy distillation: what to optimize (Section 4), where the signal comes from (Section 5), and how to stabilize training (Section 6).

This section turns to the equally important *why* and *when*: under what conditions does OPD succeed, how does it fail, what theoretical frameworks unify the diverse methods surveyed above, and when should practitioners invest in OPD versus simpler off-policy pipelines?

Crucially, these four dimensions of understanding are not independent: they form a cumulative argument. Identifying *when* OPD works (Section 7.1) reveals the preconditions that the theoretical framework must explain. Cataloging *how* it fails (Section 7.2) motivates the specific design choices, adaptive divergences, curriculum weighting, and hybrid RL objectives, whose theoretical grounding Section 7.3 provides. Both the success conditions and the failure modes then inform the practical cost-benefit analysis in Section 7.4, which distills the theoretical insights into engineering guidelines. Answering these questions is essential for moving OPD from an empirical art toward a principled engineering discipline with predictable outcomes. The analysis below draws on both theoretical frameworks (imitation learning bounds, information geometry, RL convergence theory) and systematic empirical investigations that isolate individual design choices.

7.1 Success Conditions

Li et al. (2026e) provide the most systematic investigation of OPD training dynamics, identifying two necessary conditions for effective distillation.

Thinking-pattern consistency. The student and teacher must share compatible reasoning patterns, operationalized as a high overlap ratio in their top-k token distributions. Even when the teacher achieves higher benchmark scores, a mismatched thinking pattern (e.g., a non-thinking teacher distilling into a thinking student) causes low initial overlap that training cannot fully recover. This finding implies that the teacher selection problem is not simply about capability but about *distributional compatibility*.

Genuine new knowledge. The teacher must provide capabilities beyond what the student has already acquired. When both models are trained on the same data and recipe, they converge to similar distributions at their respective scales, leaving the teacher with little transferable signal. Reverse distillation experiments show that same-family 1.5B and 7B teachers are distributionally indistinguishable from the student's perspective.

At the token level, successful OPD is characterized by progressive alignment on *high-probability overlap tokens*, a small shared token set concentrating 97–99% of the probability mass. When these conditions fail, recovery strategies include off-policy cold start (SFT initialization that reduces the pattern gap) and teacher-aligned prompt selection. Critically, the quality of OPD's dense supervision degrades systematically with trajectory depth, with instability originating at later tokens and propagating backward, raising fundamental questions about OPD's applicability to long-horizon reasoning.

These findings point toward a general principle: OPD's benefit scales with the *exploitable gap* between teacher and student. When this gap is too small (same-recipe models), the teacher offers no new information. When too large (thinking-pattern mismatch), the student cannot absorb the supervision. The productive regime lies in between, and identifying it is the central diagnostic challenge.

Diagnostic checklist. Based on the surveyed literature, we distill four pre-training diagnostics that practitioners should evaluate before committing to an OPD pipeline:

1. *Token overlap ratio* ($>$60%): Compute top-k overlap between teacher and student on a held-out set. If below 60%, apply off-policy cold start first.
2. *Pass rate on target prompts* ($>$5%): If the student cannot solve any problems in the target domain, OPD gradients will vanish. Use curriculum (PACED) or easier warm-up tasks.
3. *Teacher calibration*: Verify that the teacher's confidence correlates with correctness on student-generated prefixes. Poorly calibrated teachers inject noise.
4. *Length stability*: Monitor output length during early training. Monotonic growth indicates the KL gradient asymmetry identified by Luo et al. (2026). Apply Stable-OPD corrections preemptively.

7.2 Failure Modes

Complementing the success conditions, a growing body of work catalogs the specific ways OPD can fail *even when structural preconditions hold*, as artifacts of the on-policy training dynamics themselves. Understanding these failure modes is essential for debugging OPD pipelines in practice, and each mode has a corresponding mitigation strategy developed in the methods sections above.

We organize failure modes by their *root cause* rather than their symptom, as multiple failure modes can produce similar surface-level degradation (e.g., accuracy collapse) through different mechanisms:

The flawed prefix trap. As detailed in Section 6.1, when the student generates an erroneous prefix, the teacher's conditional distribution over subsequent tokens is poorly calibrated because the teacher was never trained on such out-of-distribution inputs. Fu et al. (2026) formally characterize three specific failure modes: (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on out-of-distribution student prefixes, and (3) tokenizer mismatch distortions. The theoretical analysis reveals a fundamental bias-variance tradeoff: token-level OPD is biased relative to sequence-level Reverse KL but enjoys tighter worst-case variance bounds.

Gradient SNR collapse. When the student's pass rate on a prompt approaches zero, all rollouts contain early errors, the teacher's signals are dominated by noise, and the gradient SNR vanishes. This is the mathematical explanation for why hard prompts yield no learning signal in standard OPD, motivating PACED's competence-boundary curriculum (Section 6.2) and HDPO's gradient augmentation (Section 5.3).

Epistemic suppression. Kim et al. (2026a) identify a subtle failure mode in self-distillation: when the process shortens reasoning traces (a common desirable outcome), it disproportionately removes *hedging phrases and uncertainty markers* that serve as implicit attention flags for difficult sub-problems. The result is shorter but less calibrated reasoning, where the model confidently commits to flawed steps it would have reconsidered in longer traces. This calibration degradation connects directly to the teacher uncertainty challenge: an overconfident student produces unreliable self-play targets, creating a vicious cycle.

Self-play saturation (the Ouroboros problem). In self-distillation settings, the student optimizes against a target sharing its own inductive biases and architectural limitations, causing the hypothesis space to progressively collapse. If the model discovers a syntactic "hack" or a highly confident but flawed reasoning heuristic, no external signal penalizes it. The model self-reinforces this flawed trajectory, driving $p_\theta(y_{\text{flawed}}|x) \to 1$. Once entirely certain of its own hallucinations, gradients vanish, exploration ceases, and the policy is trapped. This is distinct from exposure bias (which arises from train-test distribution mismatch in off-policy settings): saturation arises from the *absence* of distributional diversity in fully on-policy self-play, and is mathematically analogous to mode collapse in generative adversarial networks.

The precision-recall tradeoff. Cha & Cho (2025) formalize a fundamental tension in distillation: as the student concentrates mass on high-quality outputs (improving precision), it necessarily reduces coverage of the teacher's full output distribution (reducing recall). In the OPD setting, this manifests as the "diversity collapse" problem: aggressive Reverse KL distillation produces a student that scores highly on Pass@1 (precision) but catastrophically on Pass@k (recall), because the student has collapsed onto a single reasoning strategy per prompt. On-policy generation can *shift* this tradeoff by allowing the student to explore its own distribution rather than being constrained to teacher-demonstrated paths, potentially improving recall without sacrificing precision. Temperature-scaled sampling during on-policy generation, as used in GKD and G-OPD, provides a direct control knob: higher sampling temperatures encourage distributional exploration (increasing recall) at the cost of more noisy gradients (potentially reducing precision). This tradeoff is modulated by a single entropy-controlling parameter, providing practitioners with a principled axis for tuning distillation behavior to downstream requirements.

The calibration-capability gap. CaOPD (Zhang et al., 2026b) reveals a "Scaling Law of Miscalibration": while OPD effectively improves task accuracy, it systematically traps models in severe overconfidence. The root cause is an information mismatch: teacher supervision is formed under privileged context available during training (full reasoning traces, multiple samples), whereas the deployed model must report confidence using only its deployment-time context. By decoupling capability improvement from calibration degradation, CaOPD demonstrates that the distilled model becomes more capable but less aware of its own uncertainty boundaries. This finding has important implications for deployment: a distilled model that achieves higher benchmark scores but cannot reliably indicate when it might be wrong is arguably less safe than a weaker but better-calibrated baseline. The "illusion of certainty" connects to the saturation analysis above: both indicate that OPD can produce students that lose metacognitive capabilities (uncertainty awareness, epistemic verbalization) even as task-level performance improves.

7.3 Unified Theoretical Perspectives

The success conditions and failure modes above raise a natural question: can they be explained by a single theoretical framework? Several recent works demonstrate that they share common mathematical roots.

The unified f-divergence perspective. The framework of Wen et al. (2023) subsumes Forward KL, Reverse KL, JSD, and α-divergences as special cases parameterized by a single convexity function f. Since all these divergences can be expressed as expectations under the student's on-policy distribution, the choice of divergence is a *regularization* decision. The generator function f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$: Forward KL up-weights regions where the student underestimates the teacher (coverage), while Reverse KL up-weights regions of overestimation (precision). Wu et al. (2025) challenge the standard mode-seeking/covering dichotomy for discrete LLM distributions, demonstrating that under practical epoch budgets the difference is better understood through *head-tail focus*: Forward KL concentrates learning on the head of the teacher distribution, while Reverse KL emphasizes the tail. This ratio-weighting interpretation explains why adaptive methods that switch between divergences based on local geometry consistently outperform any fixed choice.

Why KD works: minimal explanations. Cha & Cho (2025) provide a theoretical grounding for *why* knowledge distillation works at all in generative models, identifying the minimal conditions under which a student can provably benefit from teacher supervision versus learning from data alone. Their analysis shows that the benefit of KD scales with the *information gap* between the teacher's distribution and the data distribution as seen by the student, providing a theoretical foundation for the "exploitable gap" identified empirically by Li et al. (2026e).

Unifying self-distillation and group-relative optimization. Li et al. (2026b) prove that Group-Relative Policy Optimization (GRPO) and self-distillation are mathematically equivalent under certain conditions, with the preference signal in GRPO being implicitly a form of self-distillation against the model's own best rollouts. This unification has practical implications: it suggests that the choice between GRPO-style RL and explicit self-distillation is often a matter of implementation convenience rather than fundamental algorithmic difference.

Demystifying OPD dynamics. Luo et al. (2026) address a persistent practical concern: OPD often causes *length inflation*, where the student's outputs grow progressively longer during on-policy training. They identify the root cause as an asymmetry in the KL gradient: positive KL gradients (pushing the student toward teacher-preferred tokens) are stronger than negative KL gradients (pulling away from teacher-dispreferred tokens), creating a net bias toward longer sequences. Their Stable-OPD framework introduces two complementary stabilization mechanisms: (1) a *reference divergence term* that anchors the student's length distribution to a pre-distillation baseline, preventing unconstrained growth, and (2) *rollout mixing* that blends on-policy student generations with off-policy teacher generations in a carefully tuned ratio, maintaining the benefits of on-policy training while dampening the length inflation signal. The mixing ratio decays during training: early phases use more off-policy data for stability, while later phases increase the on-policy fraction for distributional alignment. These strategies provide practical solutions to a problem that has plagued many OPD implementations and are composable with any divergence choice.

OPD as KL-constrained RL. G-OPD's equivalence proof (Section 4.3) establishes that standard OPD is a special case of dense KL-constrained RL. This bridges the distillation and RL communities, providing a shared analytical language and ensuring that advances in either field immediately transfer to the other.

Connection to preference optimization. The OPD-RL equivalence extends naturally to Direct Preference Optimization (DPO). Under DPO's closed-form solution, the optimal policy satisfies $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(r(y,x)/\beta)$. Substituting the teacher's log-probability as the implicit reward ($r(y,x) = \log P_T(y|x)$) yields $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \cdot P_T(y|x)^{1/\beta}$, which is precisely the geometric mixture distribution targeted by interpolated methods like GKD with λ-mixing. This reveals that DPO-based distillation methods (AlignDistil, ORPO-Distill) are implicitly optimizing toward the same geometric mixture as token-level KL methods, differing only in their optimization trajectory (contrastive pairs vs. direct gradient descent). The practical implication is that DPO-style methods are most useful when constructing preference pairs is cheaper than computing full teacher logits, as in API-only settings where the teacher can rank but not score.

The KD+RL gradient complementarity. The unified KD+RL framework of Li et al. (2025) decomposes the hybrid gradient into a dense KD component (stable, analytically computable) and a Monte Carlo RL component (high variance, reward-driven). The KD component dampens variance in policy gradient estimation, while the RL component prevents collapse onto suboptimal teacher modes. This theoretical insight implies that the optimal KD-to-RL weighting ratio should decrease over training as the student's policy stabilizes.

Expanding RL capabilities via text feedback. Song et al. (2026) show that textual feedback from a teacher can expand the effective capabilities of RL beyond what outcome-only rewards achieve, providing theoretical justification for the verbal feedback methods (Lion, OVD, ThinkTuning) surveyed in Section 5.2. The key insight is that text feedback provides a *denser* reward signal than binary outcomes, enabling the student to learn from partial successes that binary verification would discard.

Together, these perspectives converge on a central insight: on-policy distillation, reinforcement learning, and preference optimization are not separate training approaches but rather points on a continuous spectrum parameterized by the density of supervision, the choice of divergence, and the source of the training signal.

7.4 On-Policy vs. Off-Policy: The DeepSeek-R1 Analysis

The theoretical frameworks above establish that on-policy training offers principled advantages. But theory alone cannot settle the question of when practitioners should invest in on-policy infrastructure versus relying on simpler off-policy pipelines.

The DeepSeek-R1 anomaly. DeepSeek-R1 (DeepSeek-AI et al., 2025) represents the most prominent off-policy distillation success, training students from 1.5B to 70B on $\sim$800K samples of long chain-of-thought reasoning traces via purely off-policy SFT. On AIME 2024 (pass@1), R1-Distill-Qwen-7B reaches 55.5%, and R1-Distill-Qwen-32B reaches 72.6%. Crucially, off-policy distillation *decisively outperforms direct RL* for small-to-medium models: at the 32B scale, GRPO applied directly to Qwen2.5-32B-Base achieves only 47.0% on AIME 2024, a gap of 25.6 percentage points. This suggests that for small models, the *knowledge transfer* component of distillation (learning reasoning patterns) dominates the *distributional alignment* component (matching the student's own generation characteristics), explaining why off-policy suffices for small students but on-policy becomes necessary as student capacity grows and its generation distribution increasingly diverges from the training data.

Multi-teacher theoretical foundation. The industrial pattern of distilling from multiple specialized teachers (Qwen3, KAT-Coder-V2, Nemotron-Cascade 2) has an emerging theoretical justification. When a student distills from K specialized teachers $\{T_1, \ldots, T_K\}$ each expert in domain $\mathcal{D}_k$, the resulting student approximates a *mixture-of-experts distribution* without requiring MoE architecture. Formally, the multi-teacher objective $\sum_{k=1}^{K} w_k D_{\mathrm{KL}}(P_{T_k} \parallel P_\theta)$ produces a student that interpolates between teacher distributions, with the weights w_k controlling the relative emphasis. On-policy generation is critical here because the student must learn to *route* between reasoning strategies depending on input characteristics, a capability that static off-policy data cannot teach since the routing decision depends on the student's internal state. This explains why ORBIT's multi-budget fusion and KAT-Coder-V2's domain consolidation both require on-policy training: the *conditional strategy selection* is itself a learned capability that emerges only through on-policy exploration.

Three factors explain why off-policy works so well here: (1) the teacher is an exceptionally strong 671B MoE model whose reasoning traces cover a wide region of the problem space, providing such thorough coverage that exposure bias is empirically mitigated for most standard reasoning paths. (2) The $\sim$800K traces include diverse reasoning strategies (backtracking, verification, alternative approaches), effectively exploring multiple paths per problem. (3) The student models are relatively small (1.5B–70B), so the effective complexity of the conditional distribution (given long reasoning prefixes) is manageable for even off-policy learning.

Distillation vs. direct RL: a formal comparison. The decision between knowledge distillation and direct RL (e.g., GRPO, PPO with outcome reward) is not binary but depends on three factors: (1) *teacher quality relative to the reward signal*, where distillation is preferred when the teacher provides richer per-token information than sparse outcome rewards can encode, (2) *student scale*, where smaller students ($\leq$7B) benefit more from dense teacher supervision because their limited capacity cannot efficiently explore reward landscapes, while larger students ($\geq$32B) have sufficient capacity for RL exploration, and (3) *target ceiling*, where distillation is bounded by the teacher's capability while RL can theoretically exceed any fixed teacher through exploration. The hybrid approaches surveyed in Section 4.3 (G-OPD, KDRL, RLAD) resolve this tension by combining both: using teacher logits as a dense baseline reward that reduces variance while allowing sparse outcome rewards to drive exploration beyond the teacher's ceiling. Empirically, this combination consistently outperforms either approach alone across all student scales.

Why on-policy remains necessary. DDT (Zhang et al., 2026c) provides formal justification: the key factor driving RL's generalization advantage is on-policy data generation, which keeps training aligned with the model's evolving distribution. The off-policy ceiling, the maximum achievable performance regardless of data volume, is strictly lower because static datasets cannot cover the combinatorial space of valid reasoning paths. This ceiling gap widens with three factors: task complexity (more reasoning steps $\Rightarrow$ more paths to cover), output diversity requirements (creative tasks vs. math), and model capability (stronger students explore further from the training data).

The hybrid pipeline. Qwen3 (Yang et al., 2025a) and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) operationalize this insight, combining off-policy cold-start with on-policy refinement. This two-phase pattern, off-policy SFT to establish a strong foundation followed by on-policy OPD or RL to close the remaining distribution gap, has emerged as the standard industrial recipe. The optimal transition point (when to switch from off-policy to on-policy) depends on the student's initial capability: weaker students benefit more from extended off-policy warmup, while stronger students can transition earlier.

Compute-quality tradeoff. The expected compute cost for N tokens:

$$C_{\mathrm{off}} \approx N \times (F_{\mathrm{teacher}} + F_{\mathrm{student}} + B_{\mathrm{student}}) \tag{21}$$

$$C_{\mathrm{on}} \approx N \times (G_{\mathrm{student}} + \lambda F_{\mathrm{teacher}} + F_{\mathrm{student}} + B_{\mathrm{student}}) \tag{22}$$

where F, B denote forward and backward FLOPs, G is the autoregressive generation cost, and $\lambda \in (0, 1]$ is the teacher supervision refresh rate. Because $G_{\mathrm{student}} \gg F_{\mathrm{student}}$, on-policy training is typically 4–5$\times$ more expensive. For generic knowledge transfer, this overhead is rarely justified. For reasoning, code generation, and complex mathematics, where the off-policy ceiling is low and the on-policy marginal return is high, the investment pays off decisively.

Table 2: Comprehensive comparison of on-policy distillation methods organized by primary contribution dimension.

Method	Year	Category	Divergence/Objective	Signal	Granularity	Key Innovation
Objective: Fixed Divergences						
GKD (Agarwal et al., 2024)	2024	Objective	F-KL / R-KL / JSD	White-box	Token	Unified OPD framework, DAgger for LLMs
MiniLLM (Gu et al., 2024)	2024	Objective	Reverse KL	White-box	Sequence	REINFORCE with teacher as reward
DistiLLM (Ko et al., 2024)	2024	Objective	Skew KL (SKL+SRKL)	White-box	Token	α-smoothing for KL stability
DistiLLM-2 (Ko et al., 2025)	2025	Objective	Contrastive SKL/SRKL	White-box	Token	Asymmetric loss per data source
KETCHUP (Fan et al., 2025)	2025	Objective	Reverse KL	White-box	Sequence	K-step truncated REINFORCE
Constrained (Zimmer et al., 2025)	2025	Objective	Constrained KL	White-box	Sequence	State-augmented CMDP
Objective: Adaptive Divergences						
AKL (Wu et al., 2025)	2025	Objective	Adaptive FKL+RKL	White-box	Token	Ratio-based confidence routing
ToDi (Jung et al., 2025)	2025	Objective	Entropy-routed FKL/RKL	White-box	Token	Hard entropy threshold switching
Entropy-Aware (Jin et al., 2026)	2026	Objective	Continuous FKL+RKL blend	White-box	Token	Smooth entropy interpolation
Objective: RL-Augmented						
G-OPD (Yang et al., 2026c)	2026	Objective	KL-Constrained RL	White-box	Token/Seq	OPD $\equiv$ dense KL-RL, reward extrapolation
RLKD (Xu et al., 2025b)	2025	Objective	KD + Reward Model	White-box	Sequence	Structure-aware reward model
KDRL (Xu et al., 2025a)	2026	Objective	Joint KD + RL	White-box	Token/Seq	On-policy KL regularizer during RL
RLAD (Zhang et al., 2026g)	2026	Objective	Trust Region Ratio	White-box	Token/Seq	PPO-style selective teacher following
AlignDistil (Zhang et al., 2025a)	2025	Objective	DPO + Reverse DPO	White-box	Sequence	Synthetic preference from rollouts
SuperCorrect (Yang et al., 2025b)	2025	Objective	Template + Cross-DPO	White-box	Sequence	Hierarchical thought scaffolding
SCoRe (Lyu et al., 2025)	2025	Objective	Earliest-error correction	White-box	Sequence	Root-cause error targeting
Signal: White-Box Innovations						
DSKD (Zhang et al., 2025b)	2025	Signal	Dual-Space KL	White-box	Token	Cross-vocabulary projection
Cross-Tok. KD (Boizard et al., 2025)	2025	Signal	Latent OT alignment	White-box	Token	Optimal transport for tokenizer mismatch
Delta-KD (Ko et al., 2026)	2026	Signal	Base-to-Instruct delta	White-box	Token	Signal isolation
PromptKD (Kim et al., 2024)	2024	Signal	Prompt-based elicitation	White-box	Token	Input-side distillation
TAID (Shing et al., 2025)	2025	Dynamics	Temporal target interpolation	White-box	Token	Progressive target revelation
MiniPLM (Gu et al., 2025)	2025	Signal	Pre-training KD	White-box	Token	OPD during pre-training
Training Dynamics						
TIP (Xu et al., 2026b)	2026	Dynamics	Teacher reliability filter	White-box	Token	Prefix discrepancy attenuation
SCOPE (Zheng et al., 2026)	2026	Dynamics	Rollout routing	White-box	Token	Redirect on flawed prefix
SelecTKD (Huang et al., 2025)	2025	Dynamics	Confidence filtering	White-box	Token	Top-1 probability threshold
AdaSwitch (Peng et al., 2025)	2025	Dynamics	Exploration/guidance switch	White-box	Token	Binary mode switching
PACED (Xu et al., 2026a)	2026	Dynamics	Beta-kernel curriculum	White-box	Sample	Frontier difficulty sampling
Fast OPD (Zhang et al., 2026a)	2026	Dynamics	Prefix truncation	White-box	Hybrid	Early-token focus
Lightning OPD (Wu et al., 2026)	2026	Dynamics	Offline caching	White-box	Token	$4\times$ cost reduction
Speculative KD (Xu et al., 2025c)	2025	Dynamics	Draft-verify	White-box	Block	Amortized teacher cost
Revisiting OPD (Fu et al., 2026)	2026	Dynamics	Top-p rollout + masking	White-box	Token	Empirical failure mode fixes
Stable-OPD (Luo et al., 2026)	2026	Dynamics	R-KL + reference div.	White-box	Token	Rollout mixing for stability
Veto (Jang et al., 2026)	2026	Dynamics	Geometric bridge KL	White-box	Token	Adaptive gradient veto
Self-Distillation						
SPIN (Chen et al., 2024)	2024	Self-Play	Self-play DPO	Self	Sequence	Iterative self-vs-reference game
IRIS (Anonymous, 2026b)	2026	Self-Play	Interpolative Rényi self-play	Self	Sequence	Unifies SPIN/SPACE/SPIF
OPSD (Zhao et al., 2026b)	2026	Self (PI)	KL on GT-conditioned	Self	Token	Ground-truth as privileged info
CRISP (Sang et al., 2026)	2026	Self (PI)	Conciseness PI	Self	Token	57% token reduction, +9% accuracy
GATES (Stein et al., 2026)	2026	Self (PI)	Gated consensus KL	Self	Token	Document as PI, gated supervision
OPSDL (Anonymous, 2026a)	2026	Self (PI)	Short-ctx $\rightarrow$ long-ctx	Self	Token	Long-context self-distillation
SD-ZERO (He et al., 2026)	2026	Self (EF)	Self-revision preference	Self + Verifier	Sequence	Binary reward $\rightarrow$ dense self-supervision
π-Play (Zhang et al., 2026e)	2026	Self (PI+SP)	Multi-agent co-evolution	Self	Token	Internally generated PI from self-play
CoPD (Zhang et al., 2026f)	2026	Self (SP)	Collaborative multi-agent	Self	Sequence	Multi-agent collaborative distillation
SSD (Zhang et al., 2026d)	2026	Self (SP)	Temperature-sampled SFT	Self	Sequence	Minimalist self-play
SDPO (Hübotter et al., 2026)	2026	Self (EF)	Textual feedback + DPO	Self + Verifier	Sequence	Credit assignment from outcomes
RLTF (Song et al., 2026)	2026	Self (EF)	Critique-conditioned	Self + Verifier	Sequence	Text feedback for self-play
SRPO (Li et al., 2026b)	2026	Self (EF)	GRPO + SDPO routing	Self + Verifier	Sequence	Sample-routed multi-objective
PRISM (Li et al., 2026d)	2026	Self (SP)	Mastery-based curriculum	Self	Sequence	Progressive difficulty self-distill
PAINT (Wang et al., 2026c)	2026	Dynamics	Prefix-aligned instruction	White-box	Token	Format-aware on-policy training
Black-Box						
GAD (Ye et al., 2025)	2025	Signal	Adversarial matching	Black-box	Sequence	Discriminator as proxy reward
Lion (Jiang et al., 2023)	2023	Signal	Verbal feedback	Black-box	Sequence	NL critique curriculum
OVD (Xiong et al., 2026)	2026	Signal	Verbal scores (0–9)	Black-box	Sequence	Score-based trajectory matching
DAIL (Mendes et al., 2026)	2026	Signal	Inverse RL	Black-box	Sequence	Inferred reward from text
LUFFY (Yan et al., 2025)	2025	Signal	Mixed-policy GRPO	Off-policy + RL	Sequence	Importance-weighted off-policy
ORPO-Distill (Singh et al., 2025)	2025	Signal	Preference optimization	Black-box	Sequence	Teacher-ranked pairs

Let $\mathcal{U}(\theta, C)$ represent the student's downstream utility given compute budget C. The constrained optimization is:

$$\max \mathcal{U}(\theta, C) \quad \text{subject to } C = \gamma C_{\text{on}} + (1-\gamma) C_{\text{off}} \leq C_{\text{max}} \tag{23}$$

where γ is the fraction processed on-policy. For generic knowledge transfer, $\gamma \to 0$ suffices. For reasoning, $\frac{\partial \mathcal{U}}{\partial C_{\text{on}}} \gg \frac{\partial \mathcal{U}}{\partial C_{\text{off}}}$, justifying the compute multiplier. A natural hybrid curriculum begins with off-policy warmup ($\gamma = 0$) to establish latent representations, then progressively shifts toward $\gamma \to 1$ in later stages as the student's distribution diverges from the static data.

8 Applications, Systems, and Emerging Domains

The preceding sections developed OPD from theory through methods to unified understanding. This section examines how these algorithmic advances translate into practice. The transition from academic prototypes to industrial deployment reveals that the choice of OPD method is ultimately driven not by theoretical elegance but by practical constraints: available compute, teacher access level, the need for multi-teacher coordination, system-level

Table 3: Representative experimental configurations of OPD methods across different application domains.

Method	Teacher	Student	Training Signal	Key Benchmarks
Mathematical Reasoning				
G-OPD (Yang et al., 2026c)	Qwen2.5-32B-Instruct	Qwen2.5-Math-1.5B/7B	Dense logit + reward extrap.	MATH, GSM8K, AIME
OPSD (Zhao et al., 2026b)	Self (GT-conditioned)	Qwen2.5-Math-1.5B/7B	GT-PI self-distillation	AIME, HMMT
PACED (Xu et al., 2026a)	Qwen2.5-32B	Qwen2.5-Math-1.5B	Beta-kernel curriculum	MATH, GSM8K
KDRL (Xu et al., 2025a)	Skywork-OR1-Math-7B	R1-Distill-Qwen-1.5B	Joint KD + RL	AIME, MATH
LUFFY (Yan et al., 2025)	DeepSeek-R1 (off-policy traces)	Qwen2.5-Math-7B/1.5B	Mixed-policy GRPO	AIME, AMC, MATH-500
SCoRe (Lyu et al., 2025)	Qwen2.5-72B-Inst	Qwen2.5-7B/3B, Llama-3.1-8B	Earliest-error correction	AIME, MATH, Multi-hop QA
Instruction Following & General				
GKD (Agarwal et al., 2024)	T5-XXL (11B)	T5-Small/Base	On-policy FKL/RKL/JSD	XSum, WMT
DistiLLM (Ko et al., 2024)	GPT-2 XL (1.5B)	GPT-2 (124M)	Skewed KL	ROUGE-L, BERTScore
AlignDistil (Zhang et al., 2025a)	Qwen2.5-72B-Inst	Qwen2.5-1.5B-Inst	DPO + token-level KD	AlpacaEval, MT-Bench
Industrial Scale				
Qwen3 (Yang et al., 2025a)	Qwen3-32B/235B-A22B	Qwen3-0.6B–14B	On-policy logit distillation	AIME, MATH, LiveCode
Gemma 2 (Gemma Team et al., 2024)	Gemma 2 27B	Gemma 2 9B / 2B	KD in pre-training	MMLU, HellaSwag
MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)	Domain specialists	MiMo-V2-Flash (309B/15B MoE)	Multi-teacher logit + reward	AIME, MATH, GPQA
KAT-Coder-V2 (Li et al., 2026a)	5 domain specialists	Unified agentic coder	Specialize-then-Unify OPD	SWE-bench (79.6%)
Nemotron-Cascade 2 (Yang et al., 2026d)	Domain RL/SFT teachers	30B MoE (3B active)	Cascade RL + domain OPD	IMO, IOI, ICPC
ORBIT (Liang et al., 2026)	Multi-mode RL experts	DeepSeek-R1-1.5B, Qwen3-4B	Multi-teacher mode fusion	AIME, GPQA, MMLU-Pro
Multimodal & Domain-Specific				
VOLD (Bousselham et al., 2025)	Qwen3-8B (text-only)	Qwen2.5-VL-3B	GRPO + token-level KL	MMMU-Pro, MathVision
CORD (Hu et al., 2026)	Text-mode LALM	Audio-mode LALM	Cross-modal R-KL + GRPO	Audio QA/reasoning
Speculative KD (Xu et al., 2025c)	Gemma-7B-IT, Qwen2-7B	Gemma-2B, Qwen2-0.5B	Block-verified on-policy KL	Translation, Dialogue
Cross-Tok. KD (Boizard et al., 2025)	Llama-2-7B, Mistral-7B	OPT-350M, Pythia-160M–1B	Latent OT alignment	QA, Summarization
DP-OPD (Khadem et al., 2026)	GPT-2 Large 774M	DistilGPT-2 82M	GKD + DP-SGD (student only)	Yelp, BigPatent
Veto (Jang et al., 2026)	Qwen2-7B-IT	Qwen2-0.5B-IT	Geometric bridge KL	Reasoning, Generation
TAID (Shing et al., 2025)	GPT-2 XL, Llama-3-8B	GPT-2 Base/Med, Llama-3-2B	Temporal interpolation	Summarization, QA
GUI-SD (Chen et al., 2026)	Self (visual PI)	Qwen3-VL-Inst-8B	Entropy-guided KL	ScreenSpot, RefExp, VisGround
Self-Distillation				
SPIN (Chen et al., 2024)	Self (iter $t-1$)	Zephyr-7B-SFT	Self-play DPO	Open LLM, MT-Bench
CRISP (Sang et al., 2026)	Self (concise prompt)	Qwen3-8B/14B	Per-token R-KL	AIME, MATH-500
SD-ZERO (He et al., 2026)	Self (revision-conditioned)	Qwen3-4B-Inst, OLMo-3-7B	On-policy self-distill	AIME, MATH, Codeforces
π-Play (Zhang et al., 2026e)	Self (QCP-conditioned)	Qwen3-4B/8B	Multi-agent self-distill	NQ, TriviaQA, HotpotQA
SSD (Zhang et al., 2026d)	Self (temperature-sampled)	Qwen3-30B-Inst	On-policy SFT	LiveCodeBench v6
SDPO (Hübotter et al., 2026)	Self (textual feedback)	Qwen3-8B	Credit-assignment self-distill	Code, Math
SRPO (Li et al., 2026b)	Self (SDPO+GRPO)	Qwen3-8B	Sample-routed dual-objective	AIME, MATH, Code
Black-Box				
GAD (Ye et al., 2025)	GPT-5-Chat (API only)	Qwen2.5-14B-Inst	Adversarial minimax	LMSYS-Chat, Dolly
Lion (Jiang et al., 2023)	ChatGPT (API only)	LLaMA-7B/13B	NL critique curriculum	BIG-Bench Hard, AGIEval
OVD (Xiong et al., 2026)	QwQ-32B / Env. Agent	Qwen2.5-3B, Llama-3.2-3B	Verbal scores (0–9)	Web QA, MATH
Cross-Tok. KD (Boizard et al., 2025)	Llama-2-7B, Mistral-7B	OPT-350M, Pythia-160M–1B	Latent OT alignment	QA, Summarization

throughput bottlenecks, and domain-specific challenges (cross-modal transfer, privacy). We organize industrial applications by their *deployment pattern* and emerging domains by the new challenges they introduce.

8.1 Industrial Deployment

Recent foundation models have increasingly incorporated OPD into their post-training pipelines, revealing three distinct industrial use patterns.

Two-phase distillation pipelines. The most common industrial pattern combines off-policy cold-start with on-policy refinement. Qwen3 (Yang et al., 2025a) employs strong-to-weak distillation where larger models (32B, 235B-A22B) provide logit-level supervision on student-generated on-policy sequences, with the student generating responses in either thinking or non-thinking mode. The key architectural choice is the "mode fusion" objective: the student must learn to produce both short, direct answers (non-thinking mode) and extended chain-of-thought reasoning (thinking mode) by distilling from teachers operating in each mode independently. On-policy generation is essential here because the student's mode-selection behavior (when to think vs. when to answer directly) is itself a learned policy that cannot be captured by static datasets. Qwen3 reports that the on-policy refinement phase accounts for approximately 80% of total training compute but provides the majority of the quality improvement on reasoning benchmarks, with AIME 2024 scores improving by 15–25 absolute points over the off-policy baseline depending on student scale. Gemma 2 (Gemma Team et al., 2024) embeds knowledge distillation directly into pre-training to bridge performance gaps between parameter tiers (27B$\rightarrow$9B$\rightarrow$2B), representing a more aggressive integration where OPD is not a post-training add-on but a core training ingredient from the start. MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) combines distillation from multiple domain-specialized teachers with RL-based training, achieving frontier performance on mathematical reasoning with a 309B MoE model (15B active parameters).

OPD as model consolidation. A second pattern uses on-policy distillation for *merging* multiple specialized experts into a single deployable model. KAT-Coder-V2 (Li et al., 2026a) decomposes agentic coding into five expert domains, each undergoing independent SFT and

RL training, and consolidates them via on-policy distillation, achieving 79.6% on SWE-bench Verified. Nemotron-Cascade 2 (Yang et al., 2026d) applies the same principle to a 30B MoE model with only 3B activated parameters, achieving Gold Medal-level performance on IMO, IOI, and ICPC World Finals with 20× fewer parameters than DeepSeek-V3.2-Speciale.

OPD for multi-budget reasoning. ORBIT (Liang et al., 2026) addresses the deployment challenge of variable reasoning depth: users need different compute budgets depending on query difficulty. It produces stage-wise expert policies through RLVR under progressively tighter context budgets ($L_{k+1} = L_k/2$), then fuses them into a single model through mode-aware Reverse KL:

$$\mathcal{L}_{\text{fusion}}(\phi) = \mathbb{E}_{k\sim U(1,K)}\mathbb{E}_{\substack{q\sim\mathcal{D} \\ o\sim\pi_\phi(\cdot|q,p_k)}}\left[\sum_t \log\frac{\pi_\phi(o_t|q,p_k,o_{<t})}{\pi_{\theta_k}(o_t|q,p_k,o_{<t})}\right] \tag{24}$$

where p_k is the mode-specific prompt and $\{\pi_{\theta_k}\}$ are the frontier experts. Model merging initializes the student to mitigate the cold-start problem identified by Jang et al. (2026): without initialization near the expert manifold, the student's on-policy rollouts rarely reach the teachers' high-density regions, starving the distillation objective of useful signal. Across DeepSeek-Distill-Qwen-1.5B, Qwen3-4B, and Openmath-Nemotron-7B, ORBIT produces models that smoothly interpolate between four reasoning modes (2K to 32K token budgets) with competitive performance at each budget level, matching the initial model's accuracy at the highest budget while providing 2–7× token savings at lower budgets. The connection to the consolidation pattern above is instructive: where KAT-Coder-V2 and Nemotron-Cascade 2 merge experts specialized by *domain*, ORBIT merges experts specialized by *compute budget*, suggesting that on-policy distillation is emerging as a general-purpose multi-teacher fusion tool for any axis of model specialization.

Agentic distillation. OpenClaw-RL (Wang et al., 2026b) extends OPD to the agentic setting where the student must learn long-horizon tool use and environmental state tracking. Using hindsight-guided OPD with a Process Reward Model (PRM), it trains a Qwen3-4B agent that operates across personal, terminal, GUI, and SWE environments. Skill-SD (Wang et al., 2026a) applies importance-weighted Reverse KL with GRPO for skill-conditioned agentic learning on AppWorld and Sokoban.

These four patterns reflect a broader shift in how industry views OPD. The first pattern treats it as a *quality refinement* tool (making an existing pipeline produce better outputs). The second treats it as an *architectural simplification* tool (collapsing a complex multi-expert system into a single deployable model). The third treats it as an *inference-time control* mechanism (enabling a single model to serve multiple latency-accuracy operating points). The fourth extends it to *capability domains* where traditional training fails (agentic, multi-turn, multi-modal). The convergence of all four patterns toward on-policy generation, rather than static dataset distillation, confirms that the theoretical advantages of distributional alignment translate directly into industrial value.

8.2 Emerging Domains

Multimodal OPD. Vision-language models face unique exposure bias challenges because multimodal interleaved generation must maintain coherence across modalities, and high-quality visual reasoning data is scarce compared to text. VOLD (Bousselham et al., 2025) demonstrates that a *text-only* teacher can effectively guide a VLM student through on-policy distillation: the teacher provides reasoning supervision on student-generated visual reasoning traces without ever seeing the image, leveraging the text-based reasoning structure as a modality-agnostic scaffold. The critical finding is that an SFT cold-start phase is essential for cross-modal distributional alignment: without it, the student's on-policy rollouts in the visual reasoning space are too far from the teacher's text-based distribution for distillation to converge. On MMMU-Pro, VOLD improves Qwen2.5-VL-3B from 24.8% to 32.1%, demonstrating that cross-modal knowledge transfer through OPD can be surprisingly effective even when the teacher has no access to the visual modality.

Video-OPD (Li et al., 2026c) extends to temporal video grounding, where the student must learn to localize events within video sequences. X-OPD (Cao et al., 2026) crosses the text-speech boundary, distilling a text-mode Qwen3-Omni into its own speech mode by generating on-policy speech responses and aligning them with the model's own text-mode reasoning. CORD (Hu et al., 2026) applies cross-modal self-distillation between text and audio modes of the same model using weighted Reverse KL:

$$\mathcal{L}_{\text{CORD}} = \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{\hat{y}\sim\pi_{\text{audio}}(\cdot|x)}\left[\sum_t w_t \cdot D_{\text{KL}}(\pi_{\text{text}}(\cdot|x,\hat{y}_{<t}) \parallel \pi_{\text{audio}}(\cdot|x,\hat{y}_{<t}))\right] \tag{25}$$

where the audio mode generates on-policy rollouts and the text mode provides token-level targets, combined with GRPO for outcome reward maximization. CORD achieves state-of-the-art speech reasoning performance, demonstrating that different modality modes of the same model can serve as effective mutual teachers through on-policy interaction.

KEPO (Yang et al., 2026b) identifies a distinct failure mode in multimodal settings: sparse RLVR fails due to exploration collapse when initial solve rates are near zero. By conditioning preference optimization on teacher knowledge, KEPO provides the dense signal necessary to bootstrap capable VLMs.

Embodied intelligence. VLA-OPD (Zhong et al., 2026) applies Reverse KL on self-generated robot trajectories, distilling expert demonstrations into a Vision-Language-Action student. The action space is continuous rather than discrete, requiring the student to learn precise motor control from the teacher's dense token-level supervision on self-generated action trajectories, significantly improving sample efficiency over RL and robustness over SFT. HY-Embodied-0.5 (Tencent Robotics X and HY Vision Team, 2026) uses multi-stage OPD to compress a 32B expert VLM into a 2B embodied MoT model for spatial reasoning, outperforming similarly sized models on 16 out of 22 benchmarks.

GUI grounding. GUI-SD (Chen et al., 2026) applies on-policy self-distillation to GUI grounding, where the student generates click trajectories and the teacher (the same model with visual privileged context, including bounding box and Gaussian soft masks) provides token-level KL supervision. The entropy-guided distillation selectively weights tokens where the teacher-student divergence is highest, achieving state-of-the-art on 6 GUI grounding benchmarks with Qwen3-VL-Instruct-8B.

The consistent finding across these multimodal domains is that on-policy generation is *even more critical* in multimodal settings than in text-only ones, because the diversity of valid outputs (multiple correct grasping trajectories, multiple valid temporal localizations, multiple acceptable image descriptions) makes off-policy data coverage impractical.

Privacy-preserving OPD. DP-OPD (Khadem et al., 2026) injects differential privacy noise (DP-SGD) into the student's gradients during on-policy training while keeping the teacher frozen, proving that on-policy querying can extract utility while mathematically guaranteeing protection of the teacher's training data. This addresses a growing concern as distillation from proprietary models becomes standard practice: without privacy guarantees, on-policy querying could memorize and reproduce private training examples from the teacher.

Autonomous driving. OPD-AV (Afsharrad et al., 2026) applies GKD with $5\times$ compression for autonomous driving planning, distilling a Qwen3-8B SFT model into a 1.7B student on the nuScenes dataset.

8.3 System-Level Integration

The compute optimization methods surveyed in Section 6.3 address algorithmic efficiency, but industrial-scale OPD also requires *systems-level* solutions that span the full hardware-software stack.

Training frameworks. Production OPD pipelines must orchestrate three concurrent workloads: student rollout generation (autoregressive, memory-bound), teacher scoring (compute-bound forward pass), and student gradient update (compute-bound backward pass). Frameworks like OpenRLHF and veRL decompose these into separate process groups with distinct parallelism strategies: teacher inference uses tensor parallelism across 4–8 GPUs for low latency, student generation uses pipeline parallelism for throughput, and the optimizer uses ZeRO-3 for memory efficiency. The key engineering challenge is *synchronization*: the student's policy changes after each gradient step, invalidating all pending rollouts. Asynchronous architectures that tolerate slight policy staleness (analogous to Lightning OPD's offline caching) can achieve 30–50% higher throughput at the cost of slightly noisier gradients.

Inference serving for OPD. High-throughput serving engines (vLLM, TensorRT-LLM) designed for production inference can be repurposed for OPD's generation phase. Continuous batching and PagedAttention enable efficient generation of diverse student rollouts across a large prompt pool simultaneously, while the KV cache reuse across the shared prompt prefix reduces per-rollout memory. For white-box OPD, the serving engine must additionally expose the full vocabulary logits (rather than just the sampled token), requiring modifications to the standard speculative decoding pipeline.

Communication topology. White-box OPD with a 70B teacher and 7B student on an 8×H100 node requires transmitting the full logit tensor ($[B, T, |V|] \times 2$ bytes in BF16, e.g., $32 \times 2048 \times 128000 \times 2 \approx 16\,\text{GB}$ per batch) from the teacher to the student process group. NVLink bandwidth (900 GB/s intra-node) makes this feasible within a single node, but multi-node OPD requires careful tensor slicing to avoid saturating the inter-node InfiniBand fabric. Logit compression (top-k sparsification, quantization to INT8) can reduce this communication by 10–100× with minimal distillation quality loss.

8.4 When to Use On-Policy vs. Off-Policy

When off-policy suffices. For tasks where the target distribution is well-covered by the teacher's beam-search outputs (instruction following, factual QA), off-policy SFT on teacher-generated data provides the best cost-efficiency ratio. DeepSeek-R1 (DeepSeek-AI et al., 2025) demonstrates that off-policy distillation can achieve remarkable results even for reasoning when three conditions hold simultaneously: (1) the teacher is exceptionally strong (>500B parameters), (2) the reasoning traces include diverse strategies (backtracking, verification), and (3) the student is relatively small (≤70B) such that the conditional distribution complexity remains manageable.

When on-policy becomes necessary. The transition to on-policy distillation is justified when (1) the student exhibits clear exposure bias, performing well on teacher-style prompts but failing on user-generated ones with different phrasing, (2) the task involves multi-step reasoning where compounding errors degrade quality (Ross et al., 2011; Gudibande et al., 2023), or (3) the student needs to exceed the teacher in a specific domain via reward-guided exploration (Yang et al., 2026c). The formal criterion is whether the student's *inference-time* state visitation distribution $d_{\pi_\theta}(s)$ differs significantly from the training-time state visitation distribution $d_{\mathcal{D}}(s)$. When this divergence is large, on-policy training is mandatory.

Logit-level vs. reward-level OPD. If the teacher is white-box and the student is small (<7B), logit-level methods (GKD (Agarwal et al., 2024), DistiLLM (Ko et al., 2024)) provide the dense supervision that small models need to bootstrap competence. For larger students (≥7B) with reliable outcome verifiers, reward-guided OPD (KDRL (Xu et al., 2025a), G-OPD (Yang et al., 2026c)) enables discovery of novel solution paths beyond the teacher's capability ceiling. The hybrid approach, combining off-policy SFT warm-up with on-policy logit-level or reward-guided fine-tuning, consistently yields the best results across the surveyed literature and is the recipe adopted by all recent industrial deployments (Qwen3 (Yang et al., 2025a), MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)).

Relationship to RLHF. On-policy distillation and RLHF share identical computational structures: both generate on-policy rollouts from the student, both compute a per-trajectory signal (teacher KL vs. human preference reward), and both update the student via policy gradients or direct optimization. The distinction is purely in the signal source: RLHF uses a learned reward model trained on human preferences, while OPD uses the teacher's distribution or a verifier's binary outcome. Methods like G-OPD and REOPOLD make this equivalence formal, showing that RLHF is a special case of OPD where the "teacher" is the reward model's implicit preference distribution. This unification has practical consequences: any RLHF infrastructure (rollout generation, advantage estimation, trust regions, PPO/GRPO optimizers) can be directly repurposed for OPD with minimal modification, explaining why frameworks like OpenRLHF serve both communities.

8.5 The Distillation Tax: Compute Budget Allocation

Based on the compute analysis in Section 6.3 and the staged pipelines described in recent industrial reports (DeepSeek-AI et al., 2025; Yang et al., 2025a), a practical rule of thumb emerges for compute budget allocation. The most effective industrial pipelines follow a three-stage pattern:

1. **Off-policy warm-up** (50–70% of budget): Standard SFT on teacher-generated data provides cheap, high-bandwidth learning that establishes the student's basic capabilities and aligns its distribution close enough to the teacher for effective on-policy training.
2. **On-policy logit distillation** (20–35% of budget): Token-level OPD with adaptive divergences (GKD, DistiLLM, ToDi) closes the distribution gap on the student's own generation trajectories, addressing exposure bias where it matters most.
3. **Reward-guided refinement** (10–20% of budget): RL-augmented objectives (G-OPD, KDRL, RLAD) push the student beyond the teacher's capability ceiling on reasoning tasks, using sparse outcome rewards to discover novel solution paths.

This staged approach front-loads cheap, high-bandwidth learning and reserves expensive on-policy compute for the final quality push. Qwen3 (Yang et al., 2025a) exemplifies this pattern, with its four-stage pipeline (pre-train $\rightarrow$ SFT $\rightarrow$ RL $\rightarrow$ long-context) allocating progressively more compute to on-policy methods as training progresses. The key insight is that on-policy training is most valuable *after* off-policy warm-up has brought the student close to the teacher's distribution, minimizing the wasted rollouts that occur when the student-teacher gap is too large (Section 7.1).

9 Open Problems and Future Directions

Despite rapid progress, several fundamental challenges remain unresolved. We organize open problems by their potential impact on the field.

Distillation scaling laws. While parameter-scaling laws (Chinchilla (Hoffmann et al., 2022), Kaplan) are well-established for pre-training, no analogous framework exists for distillation. Busbridge et al. (2025) initiated the study of distillation-specific scaling, fitting parametric curves that reveal optimal teacher size grows sub-linearly with compute budget. However, the relationship between teacher-student size ratios, online rollout volume, divergence choice, and asymptotic performance remains largely unquantified. A natural conjecture is that the distillation loss follows a joint power-law:

$$L(N_S, N_T, D_{\text{on}}) = E + \frac{A}{N_S^{\alpha}} + \frac{B}{N_T^{\beta}} + \frac{C}{D_{\text{on}}^{\gamma}} + f(N_S, N_T) \tag{26}$$

where E is the irreducible task entropy and $f(N_S, N_T)$ models the capacity-gap interference. DeepSeek-R1 (DeepSeek-AI et al., 2025) provides empirical grounding: on AIME 2024, performance scales as 28.9% $\rightarrow$ 55.5% $\rightarrow$ 69.7% $\rightarrow$ 72.6% for student sizes 1.5B $\rightarrow$ 7B $\rightarrow$ 14B $\rightarrow$ 32B, with the steepest gain between 1.5B and 7B (26.6% absolute) and the 14B$\rightarrow$32B jump yielding only 2.9%. This suggests that α (student scale) exhibits rapid diminishing returns while teacher capacity dominates. Qwen3 (Yang et al., 2025a) further suggests that $f(N_S, N_T)$ is non-trivial: distilling from multiple specialized teachers yields better scaling than from a single monolithic teacher of equivalent total parameters. Establishing such scaling laws would transform OPD from an empirical art to a predictable engineering discipline.

Uncertainty-aware feedback. Teachers currently provide point-estimate probabilities. When the teacher itself is uncertain about a token, its logits carry little useful information, yet the student treats all teacher signals equally. Future OPD frameworks must incorporate the teacher's *epistemic uncertainty*, allowing the student to discount feedback on queries where the teacher is guessing. Process reward models (Lightman et al., 2024) provide a potential path: by scoring each reasoning step independently, they could identify precisely where teacher supervision is reliable versus speculative. The connection to the failure-mode analysis of Section 7.2 is direct: the flawed prefix trap is fundamentally a problem of unrecognized teacher uncertainty, and explicitly modeling this uncertainty would provide a principled solution. Concretely, one could augment the OPD objective with a teacher confidence weight $c_t = 1 - H(p_T(\cdot|x, y_{<t})) / \log|V|$, suppressing the gradient at positions where the teacher's entropy approaches maximum. TIP (Xu et al., 2026b) provides empirical support for this direction, demonstrating that teacher-student divergence-based filtering outperforms entropy-only schemes. The next step is to extend this to *predictive* uncertainty: rather than measuring the teacher's entropy on the student's current prefix, estimate what the teacher's reliability *would be* several tokens ahead, enabling the student to avoid committing to reasoning paths that will lead to regions of teacher incompetence.

Agent-level distillation. Current OPD focuses on single-turn generation. Extending OPD to multi-turn agentic settings, where the student must learn long-horizon tool use, environmental state tracking, and error recovery across entire interaction trajectories, requires fundamentally new credit assignment mechanisms. OpenClaw-RL (Wang et al., 2026b) and Skill-SD (Wang et al., 2026a) represent early steps, but the systematic degradation of teacher supervision with trajectory depth (Li et al., 2026e) poses a fundamental challenge for multi-turn distillation.

Continual and lifelong distillation. Production models undergo repeated fine-tuning cycles (safety patches, capability updates, domain adaptation). How should the distillation objective adapt when the teacher itself is being updated? SDFT (Shenfeld et al., 2026) and OEL (Ye et al., 2026a) provide initial answers for self-distillation, but the problem of distilling from a non-stationary teacher remains open.

Efficiency frontiers. The 3–8× compute overhead of on-policy training remains a barrier. While Fast OPD, Lightning OPD, and Speculative KD have made progress, the theoretical minimum cost of on-policy distillation (the information-theoretic lower bound on how much on-policy data is needed to achieve a given quality level) is unknown.

Latent-space distillation. Current OPD operates exclusively in vocabulary space, where cross-architecture alignment requires expensive projections (DSKD, Cross-Tokenizer KD). A more fundamental approach would distill in a *shared latent space* that both teacher and student can access, bypassing the vocabulary bottleneck entirely. This would require learning a common semantic manifold where distances correspond to functional similarity, enabling distillation between architecturally dissimilar models (e.g., diffusion-based text generators and autoregressive LLMs) that share no structural overlap. Recent progress in representation learning and universal sentence embeddings suggests this direction is within reach.

Cross-modal OPD theory. The extension of OPD to multimodal settings (Section 8.2) is currently driven by empirical experimentation with minimal theoretical grounding. Key open questions include: How does exposure bias manifest when multiple modalities must maintain coherence? What are the optimal divergence choices when the teacher and student operate in different modality spaces? Can the unified f-divergence framework (Section 2) be extended to handle continuous signal spaces (images, audio) alongside discrete token spaces? VOLD, X-OPD, and CORD provide empirical starting points, but the theoretical foundations for cross-modal OPD remain to be established.

Privacy-preserving OPD at scale. As surveyed in Section 8.2, DP-OPD (Khadem et al., 2026) provides the first principled framework for differentially private distillation, with an asymmetric design that applies DP-SGD only to the student while keeping the teacher frozen. The on-policy formulation is particularly well-suited to the DP setting because the student trains on its own generated prefixes, reducing the compounding errors that DP noise amplifies in autoregressive generation. Open questions include scaling DP-OPD to reasoning-capable LLMs where longer rollouts increase the privacy cost per step, and integrating DP with self-distillation methods where no separate teacher exists.

Evaluation beyond benchmarks. Standard benchmarks (MMLU, GSM8K) suffer from data contamination and measure only static pattern matching. Fu et al. (2026) systematically demonstrate that token-level OPD can appear successful on standard benchmarks while failing on distribution-shifted prompts, highlighting the gap between benchmark performance and true generalization. The precision-recall trade-off framework of Cha & Cho (2025) shows distillation induces a fundamental trade-off between sample precision and distributional coverage, modulated by entropy-controlling parameters. Future evaluation methodology should probe whether the student has learned underlying causal mechanisms rather than superficial correlations.

Closing the distillation-RL loop. Distillation and RL are historically treated as separate, linearly staged processes. Static distillation saturates when the student exhausts the teacher's knowledge, while pure RL diverges when reward signals are sparse. Chen et al. (2025) show that RL retains more pre-training capabilities than SFT because on-policy rollouts maintain distributional proximity to the student's prior. KDRL (Xu et al., 2025a) jointly optimizes KD and RL, preventing the catastrophic forgetting that plagues sequential pipelines. RE-OPOLD (Li et al., 2025) formally subsumes both as special cases, with dense KD gradients stabilizing early training and RL gradients enabling exploration. Principled methods for alternating between objectives, determining when to switch, and ensuring that RL-discovered capabilities are not overwritten during subsequent distillation phases, remain the ultimate algorithmic frontier.

Agentic OPD: open challenges. Extending OPD from single-turn text generation to multi-turn agentic settings introduces three unsolved problems. *First, environment non-stationarity*: unlike text generation where the "environment" (the prompt) is fixed, agentic environments change in response to actions, making direct trajectory comparison between teacher and student meaningless unless the teacher provides counterfactual evaluations conditioned on the student's actual state. *Second, tool-use combinatorics*: modern agent frameworks expose dozens of tools with structured argument schemas, suggesting that distillation may need to operate at the *tool-call level* rather than individual tokens, with credit assignment mechanisms that attribute trajectory success to specific tool selections. *Third, safety-critical exploration*: in coding agents, a single incorrect file write can corrupt a repository. In web-browsing agents, a misguided click can trigger irreversible actions. Future distillation frameworks must incorporate safety constraints that prevent the student from exploring catastrophically dangerous action sequences during on-policy generation, even when those sequences might be maximally informative for learning.

Continual on-policy distillation. Current OPD methods treat distillation as a discrete training phase, after which the model is deployed statically. However, the rapid pace of LLM capability advancement means that a student distilled today may benefit from re-distillation against tomorrow's stronger teacher. OEL (Ye et al., 2026a) takes a first step toward continual OPD by using interaction outcomes as dynamic PI that updates the teacher policy as the model accumulates experience. The broader challenge is designing OPD pipelines that support *lifelong distillation*: periodic re-alignment against improving teachers without catastrophic forgetting of previously acquired capabilities, without retraining from scratch, and without accumulating distributional drift across successive distillation rounds. This connects to the broader continual learning literature but adds the unique complication that both teacher and student distributions are non-stationary.

10 Conclusion

This survey has systematically examined on-policy distillation for large language models, synthesizing over 90 methods that span foundational formulations, algorithmic innovations, theoretical analyses, and industrial deployments. We organized this rapidly growing literature around a practitioner's decision chain rather than descriptive categories, tracing three fundamental design axes: *objective function* design (from fixed divergences through adaptive routing to RL-augmented objectives that break the teacher ceiling), *signal source* architecture (from dense white-box logits through black-box API access to self-distillation that eliminates the external teacher entirely), and *training dynamics* (token weighting, curriculum adaptation, and compute optimization).

Key Findings. Three central insights emerge from this synthesis. *First*, shifting from off-policy to on-policy training distributions, from $y \sim p_{\text{data}}$ to $y \sim p_\theta$, is a consistently impactful design choice, yielding meaningful accuracy gains across mathematical reasoning, code generation, and instruction following (Tables 2 and 3). The improvement stems from eliminating exposure bias: the student learns to recover from its own errors rather than memorizing idealized trajectories. Across the methods surveyed, on-policy training provides typical gains of 5–15% absolute on reasoning benchmarks over off-policy baselines, with the gap widening for longer reasoning chains. *Second*, divergence selection is not one-size-fits-all but must be adapted to both task and token position. Reverse KL excels at reasoning tasks where mode-seeking prevents hallucination, while Forward KL preserves the diversity needed for open-ended generation. Adaptive methods such as ToDi (Jung et al., 2025) and Entropy-Aware OPD (Jin et al., 2026) that dynamically select divergences based on local distributional geometry represent the current frontier. *Third*, the boundary between distillation and reinforcement learning is rapidly dissolving. G-OPD (Yang et al., 2026c), RLAD (Zhang et al., 2026g), and REOPOLD (Li et al., 2025) demonstrate that the optimal training signal combines dense teacher supervision with sparse outcome rewards, with KD providing gradient stability while RL enables exploration beyond the teacher's capability ceiling. The unification is not merely conceptual: it provides a shared toolkit (trust regions, advantage estimation, policy gradient variance reduction) that benefits both communities.

Practical Takeaways. With white-box teacher access, token-level methods (GKD (Agarwal et al., 2024), DistiLLM (Ko et al., 2024)) offer the best quality-compute tradeoff for general instruction following. For reasoning-intensive tasks, sequence-level methods (MiniLLM (Gu et al., 2024)) or hybrid approaches (Fast OPD (Zhang et al., 2026a), PACED (Xu et al., 2026a), Lightning OPD (Wu et al., 2026)) are preferable despite higher variance. Lightning OPD demonstrates that offline distillation under teacher consistency can match online OPD at 4× lower cost, significantly lowering the barrier for academic research. With only API access, GAD (Ye et al., 2025) and OVD (Xiong et al., 2026) enable effective on-policy learning through adversarial and verbal formulations, respectively. Self-distillation methods (SPIN (Chen et al., 2024), OPSD (Zhao et al., 2026b), SD-ZERO (He et al., 2026)) eliminate the need for a separate teacher entirely, making them attractive under resource constraints. The compute overhead of on-policy generation, typically 4–5× that of off-policy training, can be substantially reduced via speculative decoding (Xu et al., 2025c), prefix truncation (Zhang et al., 2026a), or offline precomputation (Wu et al., 2026).

Looking Ahead. The most pressing open problems are threefold. First, the absence of distillation scaling laws analogous to Chinchilla (Hoffmann et al., 2022) forces practitioners to rely on costly trial-and-error for compute allocation. Second, the field needs uncertainty-aware objectives that prevent the echo chamber effect when students explore out-of-distribution regions where teacher supervision is unreliable. Third, extending OPD to multi-turn agentic settings, where training involves entire interaction trajectories rather than single completions, remains a fundamental challenge compounded by the systematic degradation of teacher supervision with trajectory depth (Li et al., 2026e). As the field matures, we anticipate convergence toward unified KD+RL frameworks that treat distillation not as a separate training stage but as a continuous regularization mechanism throughout the model's lifecycle, from pre-training through deployment. As LLMs transition from static text generators to interactive reasoning agents, the principles of on-policy alignment will only grow in centrality.

Limitations. This survey focuses exclusively on methods that involve on-policy generation from the student during training. We deliberately exclude general knowledge distillation (e.g., feature-based, attention transfer), model compression techniques orthogonal to the training paradigm (pruning, quantization), and methods that use student-generated data only at inference time (self-consistency, majority voting). Our coverage is current through

April 2026. Given the field's rapid pace (approximately 8–10 new OPD papers per month), some concurrent work may be absent. Additionally, our practitioner decision tree reflects current best practices and hardware constraints. As compute costs decrease and training frameworks mature, the cost-benefit calculus will shift toward more aggressive on-policy methods.

Reproducibility challenges. A significant concern across the OPD literature is the difficulty of fair comparison. Methods are evaluated on different base models, different training compute budgets, different benchmark versions (e.g., MATH-500 vs. full MATH), and with different numbers of rollouts per prompt. Few papers control for all these variables simultaneously, making cross-method comparison unreliable from published numbers alone. Our Tables 2 and 3 report original-paper results rather than controlled reproductions, and readers should interpret cross-row comparisons with appropriate caution. We advocate for standardized OPD benchmarking protocols that fix the base model, compute budget, and evaluation suite, analogous to what HELM (Liang et al., 2023) provides for general LLM evaluation.

Broader impact. OPD has immediate implications for AI accessibility and safety.

On the accessibility front, effective distillation enables smaller, more efficient models that can run on consumer hardware, democratizing access to advanced reasoning capabilities that would otherwise require expensive API access to frontier models. The self-distillation methods surveyed here (OPSD, SD-ZERO, SSD) are particularly significant: they enable capability improvement without any external teacher, allowing resource-constrained researchers and developers to iterate rapidly.

On the safety front, OPD introduces both risks and mitigations. The risk is that effective distillation of frontier capabilities lowers barriers to misuse. The mitigation is that the teacher retains a supervisory role during training, providing a natural point for injecting safety constraints, alignment objectives, and capability boundaries. As the field matures, we expect safety-aware OPD, where the teacher not only transfers knowledge but also enforces alignment properties during the interactive training loop, to become a critical research direction.

On the environmental front, OPD's compute overhead (4–5× over off-policy SFT) raises sustainability concerns at industrial scale. However, the resulting smaller, more capable students reduce downstream inference costs by orders of magnitude, and the net carbon footprint over a model's lifetime is typically lower when amortized across billions of inference calls.

References

Amirhossein Afsharrad, Amirhesam Abedsoltan, Ahmadreza Moradipari, and Sanjay Lall. On-Policy Distillation of Language Models for Autonomous Vehicle Motion Planning. *arXiv preprint arXiv:2604.07944*, 2026. URL https://arxiv.org/abs/2604.07944.

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos, Matthieu Geist, and Olivier Bachem. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. *Proceedings of ICLR*, 2024. URL https://arxiv.org/abs/2306.13649.

Anonymous. OPSDL: On-Policy Self-Distillation for Long-Context Language Models. *arXiv preprint arXiv:2604.17535*, 2026a. URL https://arxiv.org/abs/2604.17535.

Anonymous. IRIS: Interpolative Rényi Iterative Self-play for Large Language Model Fine-Tuning. *arXiv preprint arXiv:2604.20933*, 2026b. URL https://arxiv.org/abs/2604.20933.

Nicolas Boizard, Kevin El Haddad, Céline Hudelot, and Pierre Colombo. Towards Cross-Tokenizer Distillation: the Universal Logit Distillation Loss for LLMs. *Transactions on Machine Learning Research*, 2025. URL https://arxiv.org/abs/2402.12030.

Walid Bousselham, Hilde Kuehne, and Cordelia Schmid. VOLD: Reasoning Transfer from LLMs to Vision-Language Models via On-Policy Distillation. *arXiv preprint arXiv:2510.23497*, 2025. URL https://arxiv.org/abs/2510.23497.

Dan Busbridge, Amitis Shidani, Floris Weers, Jason Ramapuram, Etai Littwin, and Russ Webb. Distillation Scaling Laws. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025. URL https://arxiv.org/abs/2502.08606.

Yuang Cai and Yuyu Yuan. $\mathcal{X}$-KD: General Experiential Knowledge Distillation for Large Language Models. *arXiv preprint arXiv:2602.12674*, 2026. URL https://arxiv.org/abs/2602.12674.

Di Cao, Dongjie Fu, Hai Yu, Siqi Zheng, Xu Tan, and Tao Jin. X-OPD: Cross-Modal On-Policy Distillation for Capability Alignment in Speech LLMs. *arXiv preprint arXiv:2603.24596*, 2026. URL https://arxiv.org/abs/2603.24596.

Sungmin Cha and Kyunghyun Cho. Why Knowledge Distillation Works in Generative Models: A Minimal Working Explanation. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL https://arxiv.org/abs/2505.13111.

Howard Chen, Noam Razin, Karthik Narasimhan, and Danqi Chen. Retaining by Doing: The Role of On-Policy Data in Mitigating Forgetting. *arXiv preprint arXiv:2510.18874*, 2025. URL https://arxiv.org/abs/2510.18874.

Kai Chen et al. Learn where to click from yourself: On-policy self-distillation for gui grounding. *arXiv preprint arXiv:2605.00642*, 2026.

Zixiang Chen, Yihe Deng, Huizhuo Yuan, Kaixuan Ji, and Quanquan Gu. Self-Play Fine-Tuning Converts Weak Language Models to Strong Language Models. *Proceedings of ICML*, 2024. URL https://arxiv.org/abs/2401.01335.

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, et al. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *Nature*, 645:633–638, 2025. URL https://arxiv.org/abs/2501.12948.

Ken Ding. HDPO: Hybrid Distillation Policy Optimization via Privileged Self-Distillation. *arXiv preprint arXiv:2603.23871*, 2026. URL https://arxiv.org/abs/2603.23871.

Jiabin Fan, Guoqing Luo, Michael Bowling, and Lili Mou. KETCHUP: K-Step Return Estimation for Sequential Knowledge Distillation. *arXiv preprint arXiv:2504.19024*, 2025. URL https://arxiv.org/abs/2504.19024.

Yuqian Fu, Haohuan Huang, Kaiwen Jiang, Yuanheng Zhu, and Dongbin Zhao. Revisiting On-Policy Distillation: Empirical Failure Modes and Simple Fixes. *arXiv preprint arXiv:2603.25562*, 2026. URL https://arxiv.org/abs/2603.25562.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, et al. Gemma 2: Improving Open Language Models at a Practical Size. *arXiv preprint arXiv:2408.00118*, 2024. URL https://arxiv.org/abs/2408.00118.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: On-Policy Distillation of Large Language Models. *Proceedings of ICLR*, 2024. URL https://arxiv.org/abs/2306.08543.

Yuxian Gu, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. MiniPLM: Knowledge Distillation for Pre-Training Language Models. *Proceedings of ICLR*, 2025. URL https://arxiv.org/abs/2410.17215.

Arnav Gudibande, Eric Wallace, Charlie Snell, Xinyang Geng, Hao Liu, Pieter Abbeel, Sergey Levine, and Dawn Song. The False Promise of Imitating Proprietary LLMs. *arXiv preprint arXiv:2305.15717*, 2023. URL https://arxiv.org/abs/2305.15717.

Yinghui He, Simran Kaur, Adithya Bhaskar, Yongjin Yang, Jiarui Liu, Narutatsu Ri, Liam Fowl, Abhishek Panigrahi, Danqi Chen, and Sanjeev Arora. Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision. *arXiv preprint arXiv:2604.12002*, 2026. URL https://arxiv.org/abs/2604.12002.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training Compute-Optimal Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://arxiv.org/abs/2203.15556.

Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling Step-by-Step! Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes. *Findings of ACL*, 2023. URL https://arxiv.org/abs/2305.02301.

Jing Hu, Danxiang Zhu, Xianlong Luo, Dan Zhang, Shuwei He, Yishu Lei, Haitao Zheng, Shikun Feng, Jingzhou He, Yu Sun, Hua Wu, and Haifeng Wang. CORD: Bridging the Audio-Text Reasoning Gap via Weighted On-policy Cross-modal Distillation. *arXiv preprint arXiv:2601.16547*, 2026. URL https://arxiv.org/abs/2601.16547.

Haiduo Huang, Jiangcheng Song, Yadong Zhang, and Pengju Ren. SelecTKD: Selective Token-Weighted Knowledge Distillation for LLMs. *arXiv preprint arXiv:2510.24021*, 2025. URL https://arxiv.org/abs/2510.24021.

Jonas Hübotter, Frederike Lübeck, Lejs Behric, Anton Baumann, Marco Bagatella, Daniel Marta, et al. Reinforcement Learning via Self-Distillation. *arXiv preprint arXiv:2601.20802*, 2026. URL https://arxiv.org/abs/2601.20802.

Ijun Jang, Jewon Yeom, Juan Yeo, Hyunggu Lim, and Taesup Kim. Stable On-Policy Distillation through Adaptive Target Reformulation. In *Findings of ACL*, 2026. URL https://arxiv.org/abs/2601.07155.

Yuxin Jiang, Chunkit Chan, Mingyang Chen, and Wei Wang. Lion: Adversarial Distillation of Proprietary Large Language Models. *Proceedings of EMNLP*, 2023. URL https://arxiv.org/abs/2305.12870.

Woogyeol Jin, Taywon Min, Yongjin Yang, Swanand Ravindra Kadhe, Yi Zhou, Dennis Wei, Nathalie Baracaldo, and Kimin Lee. Entropy-Aware On-Policy Distillation of Language Models. *arXiv preprint arXiv:2603.07079*, 2026. URL https://arxiv.org/abs/2603.07079.

Seongryong Jung, Suwan Yoon, DongGeon Kim, and Hwanhee Lee. ToDi: Token-wise Distillation via Fine-Grained Divergence Control. In *Proceedings of EMNLP*, 2025. URL https://arxiv.org/abs/2505.16297.

Fatemeh Khadem, Sajad Mousavi, Yi Fang, and Yuhong Liu. DP-OPD: Differentially Private On-Policy Distillation for Language Models. *arXiv preprint arXiv:2604.04461*, 2026. URL https://arxiv.org/abs/2604.04461.

Gyeongman Kim, Doohyuk Jang, and Eunho Yang. PromptKD: Distilling Student-Friendly Knowledge for Generative Language Models via Prompt Tuning. In *Findings of EMNLP*, 2024. URL https://arxiv.org/abs/2402.12842.

Jeonghye Kim, Xufang Luo, Minbeom Kim, Sangmook Lee, Dohyung Kim, Jiwon Jeon, Dongsheng Li, and Yuqing Yang. Why Does Self-Distillation (Sometimes) Degrade the Reasoning Capability of LLMs? *arXiv preprint arXiv:2603.24472*, 2026a. URL https://arxiv.org/abs/2603.24472.

Minsang Kim and Seung Jun Baek. Explain in Your Own Words: Improving Reasoning via Token-Selective Dual Knowledge Distillation. *Proceedings of ICLR*, 2026. URL https://arxiv.org/abs/2603.13260.

Yeongmin Kim, Donghyeok Shin, Mina Kang, Byeonghu Na, and Il-Chul Moon. Distillation of Large Language Models via Concrete Score Matching. In *International Conference on Learning Representations (ICLR)*, 2026b. URL https://arxiv.org/abs/2509.25837.

Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016a.

Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 1317–1327, 2016b.

John Kirchenbauer, Abhimanyu Hans, Brian Bartoldson, Micah Goldblum, Ashwinee Panda, and Tom Goldstein. Multi-Token Prediction via Self-Distillation. *arXiv preprint arXiv:2602.06019*, 2026. URL https://arxiv.org/abs/2602.06019.

Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. DistiLLM: Towards Streamlined Distillation for Large Language Models. *Proceedings of ICML*, 2024. URL https://arxiv.org/abs/2402.03898.

Jongwoo Ko, Tianyi Chen, Sungnyun Kim, Tianyu Ding, Luming Liang, Ilya Zharkov, and Se-Young Yun. DistiLLM-2: A Contrastive Approach Boosts the Distillation of LLMs. In *Proceedings of ICML*, 2025. URL https://arxiv.org/abs/2503.07067.

Jongwoo Ko, Sara Abdali, Young Jin Kim, Tianyi Chen, and Pashmina Cameron. Scaling Reasoning Efficiently via Relaxed On-Policy Distillation. *arXiv preprint arXiv:2603.11137*, 2026. URL https://arxiv.org/abs/2603.11137.

Fengxiang Li, Han Zhang, Haoyang Huang, et al. KAT-Coder-V2 Technical Report. *arXiv preprint arXiv:2603.27703*, 2026a. URL https://arxiv.org/abs/2603.27703.

Gengsheng Li, Tianyu Yang, Junfeng Fang, Mingyang Song, Mao Zheng, Haiyun Guo, Dan Zhang, Jinqiao Wang, and Tat-Seng Chua. Unifying Group-Relative and Self-Distillation Policy Optimization via Sample Routing. *arXiv preprint arXiv:2604.02288*, 2026b. URL https://arxiv.org/abs/2604.02288.

Jiaze Li, Hao Yin, Haoran Xu, Boshen Xu, Wenhui Tan, Zewen He, et al. Video-OPD: Efficient Post-Training of Multimodal Large Language Models for Temporal Video Grounding via On-Policy Distillation. *arXiv preprint arXiv:2602.02994*, 2026c. URL https://arxiv.org/abs/2602.02994.

Wenxuan Li et al. Prism: Progressive reasoning via iterative self-distillation with mastery-based curriculum. *arXiv preprint arXiv:2604.18892*, 2026d.

Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking On-Policy Distillation of Large Language Models: Phenomenology, Mechanism, and Recipe. *arXiv preprint arXiv:2604.13016*, 2026e. URL https://arxiv.org/abs/2604.13016.

Yingru Li, Ziniu Li, and Jiacai Liu. A Note on Hybrid Online Reinforcement and Imitation Learning for LLMs: Formulations and Algorithms. *arXiv preprint arXiv:2512.23097*, 2025. URL https://arxiv.org/abs/2512.23097.

Kun Liang, Clive Bai, Xin Xu, Chenming Tang, Sanwoo Lee, Weijie Liu, Saiyong Yang, and Yunfang Wu. ORBIT: On-policy Exploration-Exploitation for Controllable Multi-Budget Reasoning. *arXiv preprint arXiv:2601.08310*, 2026. URL https://arxiv.org/abs/2601.08310.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narang, Tatsunori Hashimoto, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's Verify Step by Step. *Proceedings of ICLR*, 2024. URL https://arxiv.org/abs/2305.20050.

Feng Luo, Yu-Neng Chuang, Guanchu Wang, Zicheng Xu, Xiaotian Han, Tianyi Zhang, and Vladimir Braverman. Demystifying OPD: Length Inflation and Stabilization Strategies for Large Language Models. *arXiv preprint arXiv:2604.08527*, 2026. URL https://arxiv.org/abs/2604.08527.

Yuanjie Lyu, Chengyu Wang, Jun Huang, and Tong Xu. From Correction to Mastery: Reinforced Distillation of Large Language Model Agents. *arXiv preprint arXiv:2509.14257*, 2025. URL https://arxiv.org/abs/2509.14257.

Ethan Mendes, Jungsoo Park, and Alan Ritter. Didactic to Constructive: Turning Expert Solutions into Learnable Reasoning. *arXiv preprint arXiv:2602.02405*, 2026. URL https://arxiv.org/abs/2602.02405.

Purbesh Mitra and Sennur Ulukus. Semantic Soft Bootstrapping: Long Context Reasoning in LLMs without Reinforcement Learning. *arXiv preprint arXiv:2512.05105*, 2025. URL https://arxiv.org/abs/2512.05105.

Emiliano Penaloza, Dheeraj Vattikonda, Nicolas Gontier, Alexandre Lacoste, Laurent Charlin, and Massimo Caccia. Privileged Information Distillation for Language Models. *arXiv preprint arXiv:2602.04942*, 2026. URL https://arxiv.org/abs/2602.04942.

Jingyu Peng, Jiaxin Mao, Yong Jiang, Zheng Liu, Pengjie Ren, Zhumin Chen, Maarten de Rijke, and Xiangyu Zhao. AdaSwitch: Balancing Exploration and Guidance in Knowledge Distillation via Adaptive Switching. *arXiv preprint arXiv:2510.07842*, 2025. URL https://arxiv.org/abs/2510.07842.

Stéphane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics*, pp. 627–635, 2011.

Aswin RRV, Jacob Dineen, Divij Handa, Md Nayem Uddin, Mihir Parmar, Chitta Baral, and Ben Zhou. ThinkTuning: Instilling Cognitive Reflections without Distillation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. URL https://arxiv.org/abs/2508.07616.

Hejian Sang, Yuanda Xu, Zhengze Zhou, Ran He, Zhipeng Wang, and Jiachen Sun. CRISP: Compressed Reasoning via Iterative Self-Policy Distillation. *arXiv preprint arXiv:2603.05433*, 2026. URL https://arxiv.org/abs/2603.05433.

Idan Shenfeld, Mehul Damani, Jonas Hübotter, and Pulkit Agrawal. Self-Distillation Enables Continual Learning. *arXiv preprint arXiv:2601.19897*, 2026. URL https://arxiv.org/abs/2601.19897.

Makoto Shing, Kou Misaki, Han Bao, Sho Yokoi, and Takuya Akiba. TAID: Temporally Adaptive Interpolated Distillation for Efficient Knowledge Transfer in Language Models. *Proceedings of ICLR*, 2025. URL https://arxiv.org/abs/2501.16937.

Aasheesh Singh, Vishal Vaddina, and Dagnachew Birru. ORPO-Distill: Mixed-Policy Preference Optimization for Cross-Architecture LLM Distillation. *NeurIPS Workshop on Efficient Reasoning*, 2025. URL https://arxiv.org/abs/2509.25100.

Yuda Song, Lili Chen, Fahim Tajwar, Remi Munos, Deepak Pathak, J. Andrew Bagnell, Aarti Singh, and Andrea Zanette. Expanding the Capabilities of Reinforcement Learning via Text Feedback. *arXiv preprint arXiv:2602.02482*, 2026. URL https://arxiv.org/abs/2602.02482.

Alex Stein, Furong Huang, and Tom Goldstein. GATES: Self-Distillation under Privileged Context with Consensus Gating. *arXiv preprint arXiv:2602.20574*, 2026. URL https://arxiv.org/abs/2602.20574.

Tencent Robotics X and HY Vision Team. HY-Embodied-0.5: Embodied Foundation Models for Real-World Agents. *arXiv preprint arXiv:2604.07430*, 2026. URL https://arxiv.org/abs/2604.07430.

Hao Wang, Guozhi Wang, Han Xiao, Yufeng Zhou, Yue Pan, Jichao Wang, Ke Xu, Yafei Wen, Xiaohu Ruan, Xiaoxin Chen, and Honggang Qi. Skill-SD: Skill-Conditioned Self-Distillation for Multi-turn LLM Agents. *arXiv preprint arXiv:2604.10674*, 2026a. URL https://arxiv.org/abs/2604.10674.

Yinjie Wang, Xuyang Chen, Xiaolong Jin, Mengdi Wang, and Ling Yang. OpenClaw-RL: Train Any Agent Simply by Talking. *arXiv preprint arXiv:2603.10165*, 2026b. URL https://arxiv.org/abs/2603.10165.

Zihao Wang et al. Paint: Prefix-aligned instruction tuning for on-policy distillation. *arXiv preprint arXiv:2604.16028*, 2026c.

Yuqiao Wen, Zichao Li, Wenyu Du, and Lili Mou. f-Divergence Minimization for Sequence-Level Knowledge Distillation. *Proceedings of ACL*, 2023. URL https://arxiv.org/abs/2307.15190.

Taiqiang Wu, Chaofan Tao, Jiahao Wang, Runming Yang, Zhe Zhao, and Ngai Wong. Rethinking Kullback-Leibler Divergence in Knowledge Distillation for Large Language Models. *Proceedings of COLING*, 2025. URL https://arxiv.org/abs/2404.02657.

Yecheng Wu, Song Han, and Han Cai. Lightning OPD: Efficient Post-Training for Large Reasoning Models with Offline On-Policy Distillation. *arXiv preprint arXiv:2604.13010*, 2026. URL https://arxiv.org/abs/2604.13010.

Xiaomi LLM-Core Team, Bangjun Xiao, Bingquan Xia, Bo Yang, Bofei Gao, Bowen Shen, Chen Zhang, Chenhong He, Chiheng Lou, Fuli Luo, Gang Wang, Gang Xie, Hailin Zhang, Hanglong Lv, et al. MiMo-V2-Flash Technical Report. *arXiv preprint arXiv:2601.02780*, 2026. URL https://arxiv.org/abs/2601.02780.

Xurong Xie, Zhucun Xue, Jiafu Wu, Jian Li, Yabiao Wang, Xiaobin Hu, et al. LLM-Oriented Token-Adaptive Knowledge Distillation. *arXiv preprint arXiv:2510.11615*, 2025. URL https://arxiv.org/abs/2510.11615.

Jing Xiong, Hui Shen, Shansan Gong, Yuxin Cheng, Jianghan Shen, Chaofan Tao, Haochen Tan, Haoli Bai, Lifeng Shang, and Ngai Wong. OVD: On-policy Verbal Distillation. *arXiv preprint arXiv:2601.21968*, 2026. URL https://arxiv.org/abs/2601.21968.

Hongling Xu, Qi Zhu, Heyuan Deng, Jinpeng Li, Lu Hou, Yasheng Wang, et al. KDRL: Post-Training Reasoning LLMs via Unified Knowledge Distillation and Reinforcement Learning. *arXiv preprint arXiv:2506.02208*, 2025a. URL https://arxiv.org/abs/2506.02208.

Shicheng Xu, Liang Pang, Yunchang Zhu, Jia Gu, Zihao Wei, Jingcheng Deng, Feiyang Pan, Huawei Shen, and Xueqi Cheng. RLKD: Distilling LLMs' Reasoning via Reinforcement Learning. *arXiv preprint arXiv:2505.16142*, 2025b. URL https://arxiv.org/abs/2505.16142.

Wenda Xu, Rujun Han, Zifeng Wang, Long T. Le, Dhruv Madeka, Lei Li, William Yang Wang, Rishabh Agarwal, Chen-Yu Lee, and Tomas Pfister. Speculative Knowledge Distillation: Bridging the Teacher-Student Gap Through Interleaved Sampling. *Proceedings of ICLR*, 2025c. URL https://arxiv.org/abs/2410.11325.

Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. A Survey on Knowledge Distillation of Large Language Models. *arXiv preprint arXiv:2402.13116*, 2024. URL https://arxiv.org/abs/2402.13116.

Yuanda Xu, Hejian Sang, Zhengze Zhou, Ran He, and Zhipeng Wang. PACED: Distillation and On-Policy Self-Distillation at the Frontier of Student Competence. *arXiv preprint arXiv:2603.11178*, 2026a. URL https://arxiv.org/abs/2603.11178.

Yuanda Xu, Hejian Sang, Zhengze Zhou, Ran He, Zhipeng Wang, and Alborz Geramifard. TIP: Token Importance in On-Policy Distillation. *arXiv preprint arXiv:2604.14084*, 2026b. URL https://arxiv.org/abs/2604.14084.

Jianhao Yan, Yafu Li, Zican Hu, Zhi Wang, Ganqu Cui, Xiaoye Qu, Yu Cheng, and Yue Zhang. Learning to Reason under Off-Policy Guidance. *arXiv preprint arXiv:2504.14945*, 2025. URL https://arxiv.org/abs/2504.14945.

Shaotian Yan, Kaiyuan Liu, Chen Shen, Bing Wang, Sinan Fan, Jun Zhang, Yue Wu, Zheng Wang, and Jieping Ye. Distribution-Aligned Sequence Distillation for Superior Long-CoT Reasoning. *arXiv preprint arXiv:2601.09088*, 2026. URL https://arxiv.org/abs/2601.09088.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, et al. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*, 2025a. URL https://arxiv.org/abs/2505.09388.

Chenxu Yang, Chuanyu Qin, Qingyi Si, Minghui Chen, Naibin Gu, Dingyu Yao, Zheng Lin, Weiping Wang, Jiaqi Wang, and Nan Duan. Self-Distilled RLVR. *arXiv preprint arXiv:2604.03128*, 2026a. URL https://arxiv.org/abs/2604.03128.

Fan Yang, Rui Meng, Trudi Di Qi, Ali Ezzati, and Yuxin Wen. KEPO: Knowledge-Enhanced Preference Optimization for Reinforcement Learning with Reasoning. *arXiv preprint arXiv:2602.00400*, 2026b. URL https://arxiv.org/abs/2602.00400.

Ling Yang, Zhaochen Yu, Tianjun Zhang, Minkai Xu, Joseph E. Gonzalez, Bin Cui, and Shuicheng Yan. SuperCorrect: Advancing Small LLM Reasoning with Thought Template Distillation and Self-Correction. *Proceedings of ICLR*, 2025b. URL https://arxiv.org/abs/2410.09008.

Wenkai Yang, Weijie Liu, Ruobing Xie, Kai Yang, Saiyong Yang, and Yankai Lin. Learning beyond Teacher: Generalized On-Policy Distillation with Reward Extrapolation. *arXiv preprint arXiv:2602.12125*, 2026c. URL https://arxiv.org/abs/2602.12125.

Zhuolin Yang, Zihan Liu, Yang Chen, Wenliang Dai, Boxin Wang, Sheng-Chieh Lin, Chankyu Lee, Yangyi Chen, Dongfu Jiang, Jiafan He, Renjie Pi, Grace Lam, Nayeon Lee, Alexander Bukharin, Mohammad Shoeybi, Bryan Catanzaro, and Wei Ping. Nemotron-Cascade 2: Post-Training LLMs with Cascade RL and Multi-Domain On-Policy Distillation. *arXiv preprint arXiv:2603.19220*, 2026d. URL https://arxiv.org/abs/2603.19220.

Tianzhu Ye, Li Dong, Zewen Chi, Xun Wu, Shaohan Huang, and Furu Wei. Black-Box On-Policy Distillation of Large Language Models. *arXiv preprint arXiv:2511.10643*, 2025. URL https://arxiv.org/abs/2511.10643.

Tianzhu Ye, Li Dong, Qingxiu Dong, Xun Wu, Shaohan Huang, and Furu Wei. Online Experiential Learning for Language Models. *arXiv preprint arXiv:2603.16856*, 2026a. URL https://arxiv.org/abs/2603.16856.

Tianzhu Ye, Li Dong, Xun Wu, Shaohan Huang, and Furu Wei. On-Policy Context Distillation for Language Models. *arXiv preprint arXiv:2602.12275*, 2026b. URL https://arxiv.org/abs/2602.12275.

Dongxu Zhang, Zhichao Yang, Sepehr Janghorbani, Jun Han, Andrew Ressler, Qian Qian, Gregory D. Lyng, Sanjit Singh Batra, and Robert E. Tillman. Fast and Effective On-policy Distillation from Reasoning Prefixes. *arXiv preprint arXiv:2602.15260*, 2026a. URL https://arxiv.org/abs/2602.15260.

Jiaxin Zhang et al. The Illusion of Certainty: Decoupling Capability and Calibration in On-Policy Distillation. *arXiv preprint arXiv:2604.16830*, 2026b. URL https://arxiv.org/abs/2604.16830.

Miaosen Zhang, Yishan Liu, Shuxia Lin, Xu Yang, Qi Dai, Chong Luo, Weihao Jiang, Peng Hou, Anxiang Zeng, Xin Geng, and Baining Guo. Towards On-Policy SFT: Distribution Discriminant Theory and its Applications in LLM Training. *arXiv preprint arXiv:2602.12222*, 2026c. URL https://arxiv.org/abs/2602.12222.

Ruixiang Zhang, Richard He Bai, Huangjie Zheng, Navdeep Jaitly, Ronan Collobert, and Yizhe Zhang. Embarrassingly Simple Self-Distillation Improves Code Generation. *arXiv preprint arXiv:2604.01193*, 2026d. URL https://arxiv.org/abs/2604.01193.

Songming Zhang, Xue Zhang, Tong Zhang, Bojie Hu, Yufeng Chen, and Jinan Xu. AlignDistil: Token-Level Language Model Alignment as Adaptive Policy Distillation. In *Proceedings of ACL*, 2025a. URL https://arxiv.org/abs/2503.02832.

Xue Zhang, Songming Zhang, Yunlong Liang, Fandong Meng, Yufeng Chen, Jinan Xu, and Jie Zhou. A Dual-Space Framework for General Knowledge Distillation of Large Language Models. *arXiv preprint arXiv:2504.11426*, 2025b. URL https://arxiv.org/abs/2504.11426.

Yaocheng Zhang, Yuanheng Zhu, Wenyue Chong, Songjun Tu, Qichao Zhang, Jiajun Chai, Xiaohan Wang, Wei Lin, Guojun Yin, and Dongbin Zhao. π-Play: Multi-Agent Self-Play via Privileged Self-Distillation without External Data. *arXiv preprint arXiv:2604.14054*, 2026e. URL https://arxiv.org/abs/2604.14054.

Yuxiang Zhang et al. Copd: Collaborative on-policy distillation with multi-agent self-play. *arXiv preprint arXiv:2605.00123*, 2026f.

Zhaoyang Zhang, Shuli Jiang, Yantao Shen, Yuting Zhang, Dhananjay Ram, Shuo Yang, et al. Reinforcement-aware Knowledge Distillation for LLM Reasoning. *arXiv preprint arXiv:2602.22495*, 2026g. URL https://arxiv.org/abs/2602.22495.

Anhao Zhao, Ziyang Chen, Junlong Tong, Yingqi Fan, Fanghua Ye, Shuhao Li, Yunpu Ma, Wenjie Li, and Xiaoyu Shen. On-Policy Supervised Fine-Tuning for Efficient Reasoning. *arXiv preprint arXiv:2602.13407*, 2026a. URL https://arxiv.org/abs/2602.13407.

Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models. *arXiv preprint arXiv:2601.18734*, 2026b. URL https://arxiv.org/abs/2601.18734.

Binbin Zheng, Xing Ma, Yiheng Liang, Jingqing Ruan, Xiaoliang Fu, Kepeng Lin, Benchang Zhu, Ke Zeng, and Xunliang Cai. SCOPE: Signal-Calibrated On-Policy Distillation Enhancement with Dual-Path Adaptive Weighting. *arXiv preprint arXiv:2604.10688*, 2026. URL https://arxiv.org/abs/2604.10688.

Qihuang Zhong, Liang Ding, Li Shen, Juhua Liu, Bo Du, and Dacheng Tao. Revisiting Knowledge Distillation for Autoregressive Language Models. *Proceedings of ACL*, 2024. URL https://arxiv.org/abs/2402.11890.

Zhide Zhong, Haodong Yan, Junfeng Li, Junjie He, Tianran Zhang, and Haoang Li. VLA-OPD: Bridging Offline SFT and Online RL for Vision-Language-Action Models via On-Policy Distillation. *arXiv preprint arXiv:2603.26666*, 2026. URL https://arxiv.org/abs/2603.26666.

Yongchao Zhou, Kaifeng Lyu, Ankit Singh Rawat, Aditya Krishna Menon, Afshin Rostamizadeh, Sanjiv Kumar, Jean-François Kagy, and Rishabh Agarwal. DistillSpec: Improving Speculative Decoding via Knowledge Distillation. *arXiv preprint arXiv:2310.08461*, 2023. URL https://arxiv.org/abs/2310.08461.

Matthieu Zimmer, Xiaotong Ji, Tu Nguyen, and Haitham Bou Ammar. Rethinking Large Language Model Distillation: A Constrained Markov Decision Process Perspective. *arXiv preprint arXiv:2509.22921*, 2025. URL https://arxiv.org/abs/2509.22921.